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Foreword and acknowledgements

TBC...

This work, analysis and report have been a team effort. The following people have made a material contribution to all or part of the development of this work and the various outputs:

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New Hospitals Programme - Cheryl Riotto, Anne White, Joanna Wood, Carolyn Akyil, Dr. Michael Weinbren.

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Abstract

Objectives: Traditionally, hospital beds have been in multi-bed bays, however, there has been a move towards single occupancy inpatient rooms. Numerical evidence on the impact of single rooms is limited. We sought to quantify the potential impacts of switching to single-bed accommodation to help inform the planning and design of future hospitals.

Design & setting: We used causal impact analysis to quantify the change in productivity and efficiency measures (length of stay, waiting time, bed occupancy, emergency readmissions, cleaning costs), health and safety measures (in-hospital deaths, in-hospital falls and injuries, hospital acquired c.difficile), and patient and staff experience (Friends and Family Test for inpatient experience, staff survey, staff sickness and staff turnover) in 7 study hospitals in the English National Health Service that switched to entirely or majority single-bed rooms between 2010 to 2022, with the use of comparable control hospitals.

Results: There was significant variability between hospitals following switching to single rooms. For bed occupancy, emergency readmissions, cleaning costs and all health and safety measures there were either positive effects or no change. However, positive, negative and no effects were seen for length of stay, waiting time, patient experience, and staff experience measures. Reduced length of stay could help offset additional costs associated with single room hospitals, however, at general acute hospitals there was no significant change in length of stay (7% reduction (-17% to 2% 95% CI)). At general acute hospitals the proportion of staff recommending their trust as a place to work significantly decreased by 15% (-23% to -6% 95% CI) and staff sickness absences increased by 6% (1% to 12% 95% CI).

Conclusion: There was significant variability within measures and between hospitals. Most indicator measures showed no consistent or significant change following a switch to single rooms, although staff do appear to be

negatively affected in some ways. These findings are broadly in line with the published literature and also feedback from an independent listening exercise with representatives from these hospitals.

Introduction and background

Since its inception, the New Hospital Programme (NHP) has advocated for single bed rooms (SBR). The grounds for this are based on a range of guidance, evidence and previous policy statements of intent. Whilst that position is unequivocal, there is uncertainty around the quantification of impacts of such a change and thus the assumptions that feed into demand and capacity models and economic business cases for current and future new hospital developments.

Specifically, a starting assumption was that a length of stay reduction of 12% was likely for all new SBR facilities compared to prior trends. This assumption was criticised by the national audit office (NAO) ([National Audit Office 2023](#)) in the face of a mixed evidence base.

It is the express aim of this analysis therefore to quantify the range of likely impacts for length of stay and a basket of other measures across patient & staff experience, productivity & efficiency and health & safety. Where there are differential impacts, i.e. for specialist units compared to general acute, these will also be highlighted to aid future planning assumptions.

Executive Summary

Traditionally, hospital beds have been in shared multi-bed bays, however, there has been a move both in the UK and globally towards single-bed inpatient rooms. This analysis was undertaken to better understand the impact of switching to single-bed accommodation when building future hospitals.

There are 7 hospitals in England where single-bed inpatient accommodation replaced multi-bed accommodation fully or in part. We have used these hospital sites as natural experiments, and with the use of matched control sites are able to quantify changes in these study sites following their switch to single bedrooms.

Our analysis investigates the impact on measures from three domains 1) Health and Safety, 2) Patient and Staff Experience and 3) Productivity and Efficiency to determine the extent of any benefits, or negative impacts that need to be mitigated with single-bed room sites.

Our results are not consistent across hospital sites, suggesting that there are a number of other factors in the hospital design and procedures that are important, hence switching to single bedroom hospitals without considering the other factors would not necessarily have a positive effect on outcomes.

For productivity and efficiency measures there is a general although not exclusive trend towards positive impacts following a switch to single bed rooms. While overall at general acute hospitals there are no significant changes, the data trends towards a decrease in mean length of stay and rate of emergency readmissions and an increase in bed occupancy.

Switching to single bedrooms appears to have a positive or neutral effect on Health and Safety measures. It could be considered that no change in health and safety measures is a positive effect, as patient safety is not being negatively impacted. While data is limited there is a tendency towards decreased C. difficile infection rates.

In terms of patient and staff experiences, the switch to single bedrooms tends to result in some negative effects for staff. For general acute hospitals there was a significant reduction of 15% in the percentage of staff who would recommend their trust as a place to work and a significant 6% increase in staff sickness. While this might, at least in part, be related to the move to a new hospital site that is less convenient for staff, it will be important for staff views and experiences should be considered when designing hospitals with single bedrooms and thought given to how potentially detrimental effects on staff may be mitigated. The percentage of patients reporting a positive experience increases by a small, but statistically significant 1%.

What we already know

There have been 2 NHP commissioned pieces of work in the recent past aiming to better understand the nature and scale of any impacts related to switching from multi-occupancy hospital rooms to single-bed rooms.

The Health Economics Unit undertook a literature review ([Health Economics Unit 2022](#)) primarily around clinical impacts, patient/staff experience and the economic impacts of the structural changes. Using evidence from the literature the review highlighted the following factors that may potentially be impacted in some way:

- Mortality
- Complications of disease
- Healthcare Acquired Infections (HCAI)
- Falls and patient safety
- Readmissions
- Privacy
- Loneliness
- Disturbance
- Quality of Life
- Length of stay
- Costs of care

The conclusions of the review were that **“moving adult and child inpatients from older hospitals with largely shared rooms to fully single-room accommodation is unlikely to lead to a meaningful difference in clinical outcomes, may reduce overall costs per patient, and will lead to greater satisfaction in a majority of patients. However, switching to 100% single rooms will leave a substantial minority of people feeling lonely and socially isolated, with many patients preferring to be in a shared room. A compromise approach – where accommodation is largely in single rooms but includes some bays with two to four beds – may provide maximum satisfaction and minimise overall costs without adversely affecting clinical outcomes.”**

More latterly, the Innovations team at NHP have created a living evidence review ([Wood 2024](#)) on the topic to identify new and emerging evidence of effects of implementing single bedrooms and their potential scale. The review highlighted the ongoing variable nature of impacts with both positive, negative and neutral impacts found across length of stay, mortality, patient experience and staff attitudes.

Broadly speaking, the evidence points towards an increase in falls/fall risk, improved infection control and increased patient satisfaction with admitted care. More positive outcome impacts were generally demonstrated for specific clinical subgroups of patients notably intensive care; maternal and neonates; oncology.

The review summary suggests that **“Overall, 87% of studies analysed by Taylor et al., 2018 reported advantages associated with single patient rooms. Of these, 27% reported a combination of advantages and disadvantages, while 22% of the studies reported a combination of advantages and neutral results.”**

Hospital Background

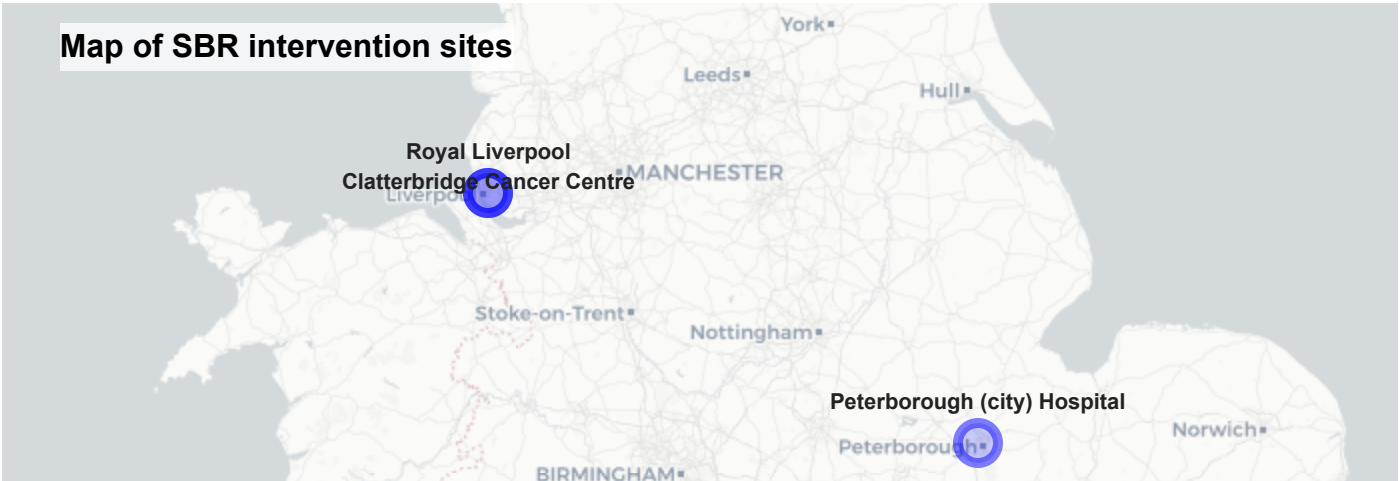
Initially 11 hospitals were identified to have made a switch to single bedrooms. However due to being unable to access data for hospitals outside England, three hospitals (Ulster Hospital, Glasgow Central and The Grange Hospital) were excluded from this analysis. James Paget hospital made a switch to single bedrooms in May 2023 but was also excluded from analysis, due to the switch being too recent and that currently the ward is used as interim beds for RAAC remedial work.

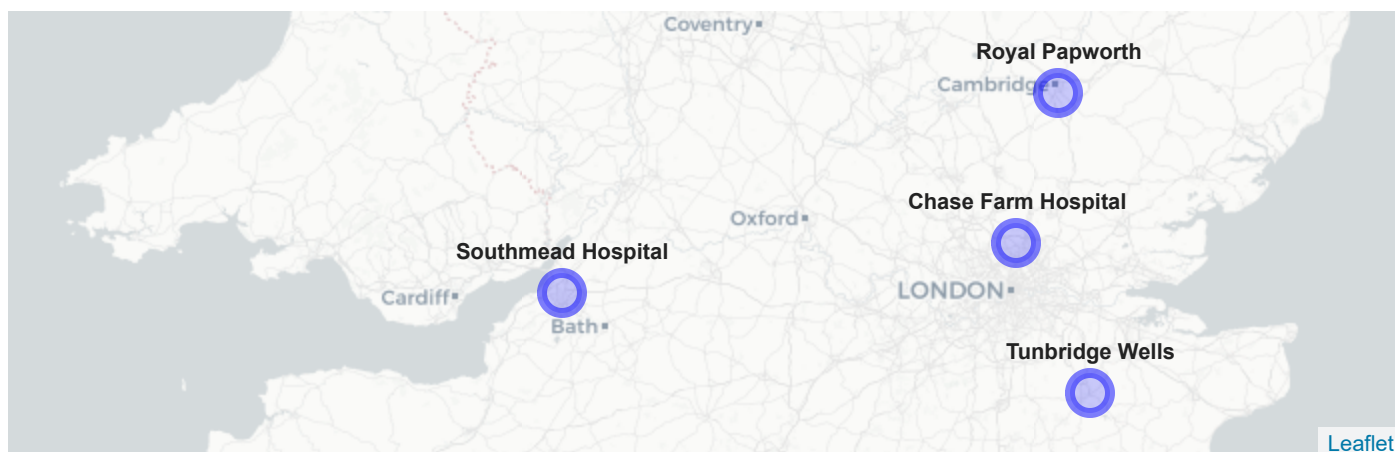
An overview of the remaining seven hospitals that switched to SBR is shown in [Table 1](#). Data on the percentage of single bedrooms before and after the switch at each site was not readily available or inconsistent, so an indication of whether the switch was partial or full is shown. Plots of the percentage of single bedrooms at provider level calculated from two different data submissions are available in Appendix E ([Section 12.5](#)).

Table 1:
Details of the switch to single bed accommodation at the seven intervention sites.

Site	Provider	Switch month	Type of switch
Royal Liverpool	Liverpool University Hospitals NHS Foundation Trust	Oct 2022	full
Clatterbridge Cancer Centre	The Clatterbridge Cancer Centre NHS Foundation Trust	Jun 2020	full
Royal Papworth	Royal Papworth Hospital NHS Foundation Trust	May 2019	full
Chase Farm Hospital	Royal Free London NHS Foundation Trust	Sep 2018	partial
Southmead Hospital	North Bristol NHS Trust	May 2014	partial
Tunbridge Wells	Maidstone and Tunbridge Wells NHS Trust	Jan 2011	full
Peterborough (city) Hospital	North West Anglia NHS Foundation Trust	Nov 2010	partial

n: 7





Location of 7 natural experiment sites in England

Royal Liverpool University Hospital

Royal Liverpool University Hospital is a large acute hospital. It mainly focuses on complex planned care and specialist services including nephrology, renal transplantation, nuclear medicine, haematology, lithotripsy, dermatology and urology ([Liverpool University Hospitals 2024](#)). The provider Liverpool University Hospitals NHS Foundation also manages Aintree University hospital and Broadgreen hospital. Royal Liverpool University Hospital moved to a new site with fully single bed accommodation in October 2022.

Clatterbridge Cancer Centre

The Clatterbridge Trust operates 3 cancer centres in the Cheshire and Merseyside area. The Liverpool centre moved to fully single-bed rooms in June 2020. The centre offers the full range of cancer treatment, both routine and surgical as well as diagnostics, outpatient appointments and associated support services ([Clatterbridge Cancer NHS Trust 2024](#)).

Royal Papworth

Royal Papworth hospital is a specialist centre for cardiothoracic (heart and lung) treatment, surgery and transplantation ([Royal Papworth Hospital 2024](#)). The move to fully single bed accommodation occurred in May 2019.

Chase Farm

Chase Farm hospital is a planned care centre operated by the Royal Free London NHS Trust. The Trust also runs the Royal Free and Barnet Hospital sites as well as other acute service across greater London ([Royal Free London 2024](#)). The new-build Chase Farm site, opened in September 2018 with a total of 50 beds, 42 were single bed rooms ([Flint 2018](#)).

Southmead

Southmead hospital is run by the North Bristol NHS Trust who also operate Frenchay and Cossham Hospitals as well as other acute and rehabilitation services. The Brunel building housing the hospital opened in May 2014 with three quarters of the beds in single rooms and the remainder in four-bed bays ([North Bristol NHS Trust 2024](#)).

Tunbridge Wells

The new hospital at Tunbridge Wells, replacing services at Pembury and Kent & Sussex, opened in two phases in January 2011 (maternity and children) and September 2011 (other inpatient services). Tunbridge Wells was the first hospital in England to switch to fully single bedroom accommodation. ([Jill Maben 2014](#))

Peterborough

Peterborough City Hospital, operated by North West Anglia NHS Trust, was opened in November 2010 offering a range of acute and specialist services ([North West Anglia 2024](#)). The site opened with a mix of single bedrooms and three- to four-bedded ward areas.

Impact measures and data sources

The evidence suggests that impacts can be broadly categorised into three domains: health & safety, staff & patient experience and productivity & efficiency. A long list of potential impacts were drawn from the existing evidence reviews and measures were sought from published data that were specific to the logic/causal relationship or were proxies for the suggested effect. Measures were included within the study if there were sufficient time series, if the data were consistent and if the data were available at least at provider level.

Where existing data sources were lacking, and for those impacts based on hospital data, we derived suitable measures from Secondary User Service (SUS) data held within NCDR ([Arden GEM CSU, n.d.](#)).

Our final set of impact measures are thus:

Table 2:

Information on indicator source, provenance, derivation and rationale.

id	domain	name	desc	source	derivation_note
1	Health and Safety	Hospital deaths	Rate of death per 1,000 discharges.	Derived from SUS admitted patient care data (NCDR)	Ordinary inpatient only. Stillbirth excluded. NHS providers only.

id	domain	name	desc	source	derivation_note
2	Health and Safety	Healthcare Associated Infections - clostridium difficile	Cases per 10,000 bed days	https://www.gov.uk/government/collections/healthcare-associated-infections-hcai-guidance-data-and-analysis	Bed days denominator derived from SU: APCE
3	Health and Safety	Falls and injuries	Spells with fall as % of all spells	Derived from SUS admitted patient care data (NCDR)	Distinct spells where episode order in spell > 1 AND ICD-10 code in (W01, W03-W08, W18-W19, S0-S9, T0, T10-T14) where non present in first episode.
4	Productivity and Efficiency	Length of Stay	Mean length of stay	Derived from SUS admitted patient care data (NCDR)	Ordinary inpatient only, deaths excluded, completed spells only, NHS providers only.

id	domain	name	desc	source	derivation_note
5	Productivity and Efficiency	Waiting time	Median wait time in weeks	https://www.england.nhs.uk/statistics/statistical-work-areas/rtt-waiting-times	
6	Productivity and Efficiency	Emergency readmissions	% of discharges with emergency readmission < 30 days	Derived from SUS admitted patient care data (NCDR)	Ordinary inpatient only, discharge method = 1 or 3
7	Productivity and Efficiency	Bed occupancy	% available beds occupied overnight	https://www.england.nhs.uk/statistics/statistical-work-areas/bed-availability-and-occupancy/	

id	domain	name	desc	source	derivation_note
8	Productivity and Efficiency	Cleaning costs	Total cleaning costs (£/sqm of occupied floor space)	https://digital.nhs.uk/data-and-information/publications/statistical/estates-returns-information-collection	
9	Patient and Staff Experience	Patient experience	Family and friends test, % positive scores (inpatient)	https://www.england.nhs.uk/fft/friends-and-family-test-data/	
10	Patient and Staff Experience	Staff sickness	% available days taken off sick	https://digital.nhs.uk/data-and-information/publications/statistical/nhs-sickness-absence-rates/august-2023-provisional-statistics	

id	domain	name	desc	source	derivation_note
11	Patient and Staff Experience	Staff turnover	% of staff leaving	https://digital.nhs.uk/data-and-information/publications/statistical/nhs-workforce-statistics	
12	Patient and Staff Experience	Staff experience	% staff recommending the trust as place to work	https://www.nhsstaffsurveys.com/results/	
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Availability of indicators over time and for each site

[Figure 1](#) demonstrates the availability of indicator datasets over time and the month of the switch to SBR for the seven study sites to allow for general assessment of suitability for testing.

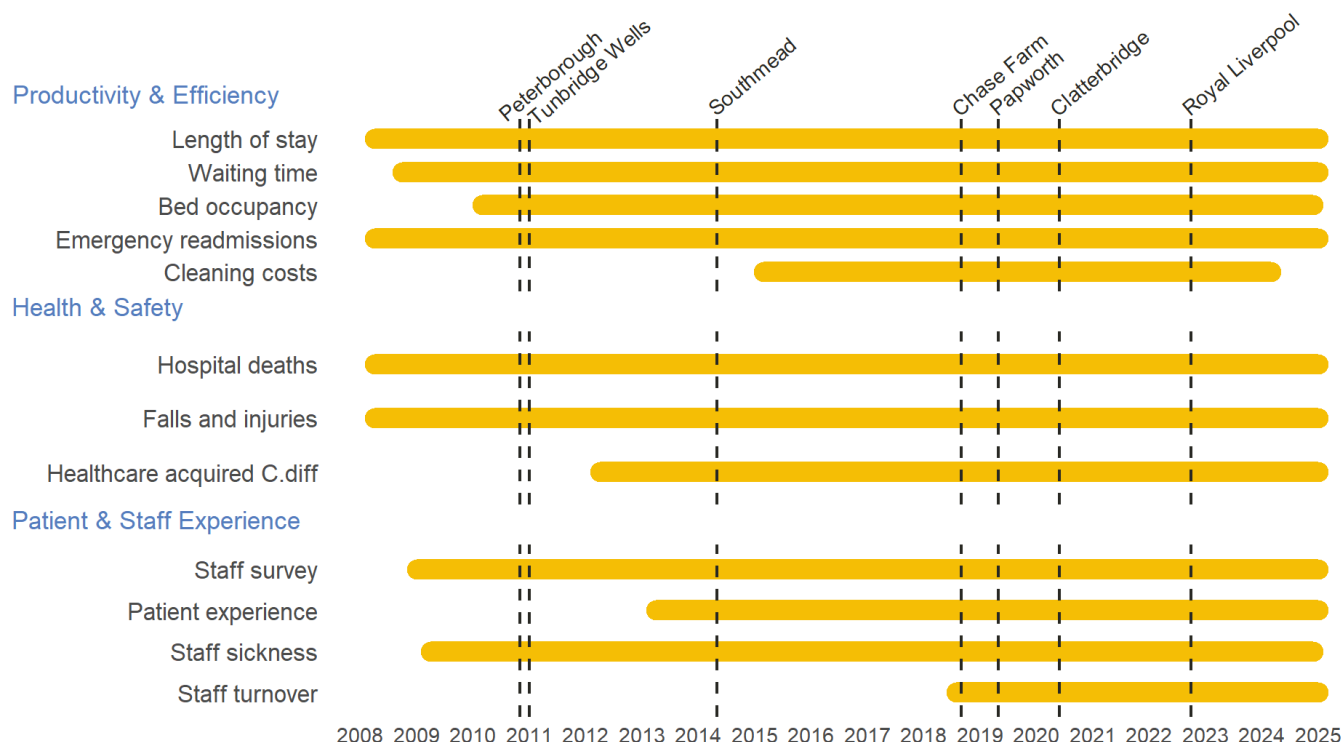


Figure 1: Availability of indicators over time. Dashed lines represent the month of the switch to SBR in each intervention site.

Additional data issues have been uncovered during preparation of the data for analysis and some further omissions of some indicator/site combinations have been made. Full notes on data availability and suitability in relation to the analysis can be found in Appendix B ([Section 12.2](#)).

[Table 3](#) summarises which indicators have been investigated for each site, and where analyses were not possible due to lack of data, other data issues or inability to fit a suitable model to the data.

Table 3:

Summary of indicators that were investigated for each site.

Variable	Level	Royal Liverpool	Clatterbridge	Papworth	Chase Farm	Southmead	Tunbridge Wells	Peterborough
Productivity & Efficiency								
Length of stay	Site	✓	✓	-	✓	✓	✓	✓
Waiting time	Provider	✓	✓	✓	-	-	✓	✓
Bed occupancy	Provider	✓	✓	-	✓	✓	x	x
Emergency readmissions	Site	✓	✓	✓	✓	✓	✓	✓
Cleaning costs	Site	✓	✓	✓	-	x	x	x

✓ indicates there was sufficient/suitable data to conduct causal impact analysis. 'x' indicates that there was not sufficient time series data available to analyse. '-' indicates data issues that meant it was not possible to conduct causal impact analysis.

Variable	Level	Royal Liverpool	Clatterbridge	Papworth	Chase Farm	Southmead	Tunbridge Wells	Peterborough
Health & Safety								
Hospital deaths	Site	✓	✓	-	-	✓	✓	✓
Falls and injuries	Site	✓	-	-	-	✓	✓	✓
Healthcare acquired C.diff	Provider	✓	-	-	✓	✓	x	x
Patient & Staff Experience								
Patient experience	Site	✓	✓	✓	✓	✓	x	x
Staff sickness	Provider	✓	✓	✓	✓	✓	✓	✓
Staff turnover	Provider	✓	✓	✓	x	x	x	x
Staff survey	Provider	✓	✓	✓	✓	✓	✓	x

✓ indicates there was sufficient/suitable data to conduct causal impact analysis.'x' indicates that there was not sufficient time series data available to analyse.'-' indicates data issues that meant it was not possible to conduct causal impact analysis.

For some sites (i.e Chase Farm, Clatterbridge and Papworth) at least part of the outcome period is impacted by COVID-19. While COVID-19 effects will also be present in the matched control sites, the magnitude of effects varies greatly between hospitals. Indeed, for some measures Papworth was excluded as it appears to have been impacted to a much greater degree by COVID-19 than its control sites.

Where there is missing data or anomalies in the outcome periods, i.e. sudden increases or decreases, the outcome period has been cut short. For the Health and Safety indicators the rates of these adverse events were generally low at the specialist sites (Papworth, Clatterbridge and Chase Farm), often dropping to zero in some months, which meant it was not always possible to fit a suitable causal impact model.

Some data is only available at a provider level, this means that where the switch to single bedrooms occurred at a provider with multiple sites the effects of the switch will be diluted by the lack of the change in single bedroom percentage at the providers other sites.

Description of methods

Causal Impact Analysis is a technique that uses a control group to estimate the post-intervention counterfactual against which to measure study group impacts ([Broderick K 2015](#)). The relationship between the control group (similar hospital providers, patient case-mix and prior trends) and the study site pre-switch is determined and used along with the control sites data post-switch to calculate a synthetic baseline that predicts what would have happened to the outcome of interest at the study site had there been no switch. We can then compare this synthetic baseline with the actual outcome at the site of interest to quantify the change in the study hospitals following their switch to single-bed rooms.

Finding control hospitals

There was a multi-phase selection process to create a pool of controls for each study site. A summary of these steps is presented in [Figure 2](#). The first step was to match the study site and controls on type of site. For Clatterbridge the pool was limited to cancer-only sites and for Royal Papworth the pool was limited to cardiac sites. For all other study sites the pool was limited to general acute sites.

Controls were then matched on the percentage of single bed rooms available at the provider prior to the switch. Providers were excluded if the difference between their SBR percentage and the study's provider was greater than 5% in the month before the switch to SBR. Then providers were excluded if their SBR percentage increased by 5% or more after this point.

Differences in floor space, the median age of inpatients and the elective:non-elective ratio of admissions were calculated between the controls and study sites and ranked. A similarity rank was calculated by ranking the sum of these three ranks and sites with a similarity rank less than or equal to 20 formed the control pool for further analysis. Due to Clatterbridge and Royal Papworth being specialist trusts, there were only 4 and 9 sites respectively in the control pools.

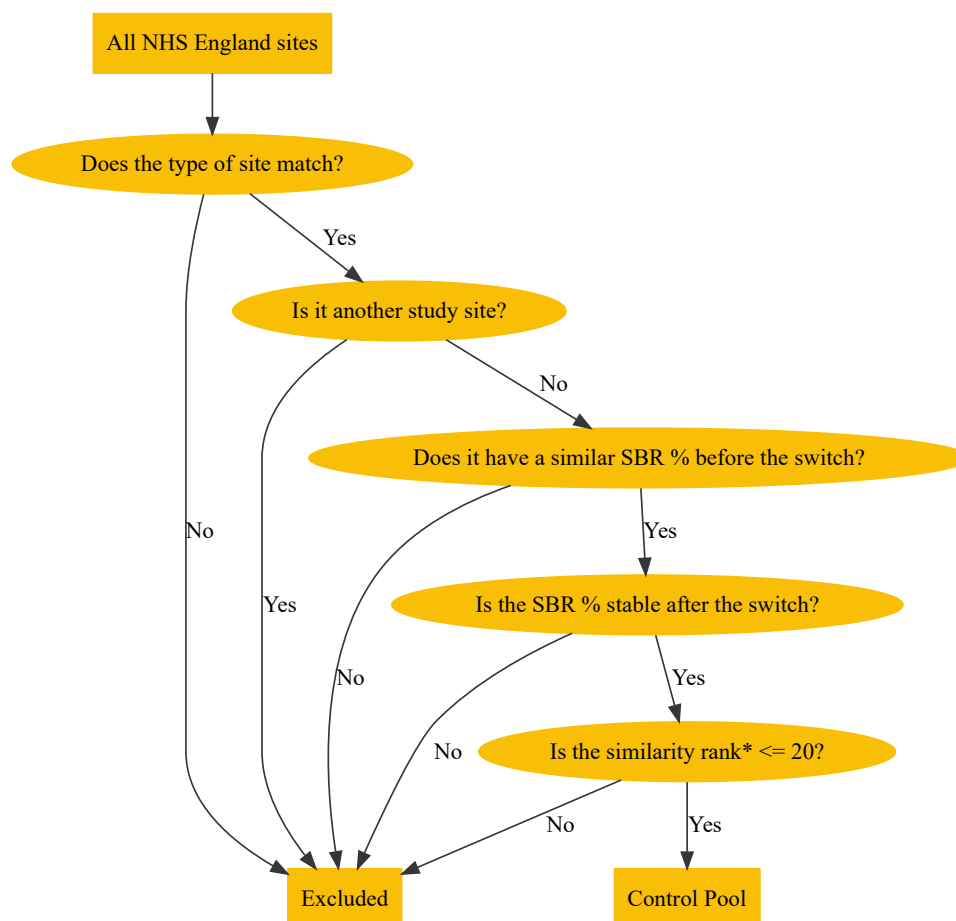


Figure 2: Flow chart of the multi-phase process used to select the top 20 controls for each site. *The similarity rank is calculated as the sum of the ranked difference of floor space, median age of patients and the elective:non-elective ratio of admissions.

A full list and map of the sites in the control pools for each study site can be found in Appendix C ([Section 12.3](#)).

Causal Impact Analysis

The causal impact analysis is implemented using the `MarketMatching` (Larsen 2024) package in R, which uses the `dtw` (Giorgino 2022) and `CausalImpact` (Brodersen and Hauser 2014) R packages to provide a simple way to find the most appropriate control matches using dynamic time warping and then using these controls for causal impact analysis using Bayesian structural time-series models. Using control sites means that seasonal changes are accounted for within the analysis. The dynamic time warping approach prevents potential controls being excluded if their relationship with the study site is temporarily shifted; it accounts for any shifts in any historic trends, such as a winter peak starting slightly earlier or later at a potential control site.

Following the multi-phase control selection process outlined in [Section 8.1](#), the control pools for each site were fed into the `MarketMatching` package. This step selects the best matched control sites based on them having similar trends in the variable of interest over time to the study site. Up to 5 control sites were selected for each study site, but for some sites, particularly the specialist centres, the number of appropriate matches was smaller and so fewer than 5 control sites were selected. Control sites are selected for each site and indicator variable and as this matching of controls is based on similar trends over time a study site may have different control sites selected for different indicators. A list of the control providers/sites identified by `MarketMatching` for each indicator can be found in Appendix D ([Section 12.4](#)).

The causal impact models were generated for each site and for each indicator where sufficient data was available to fit a model and there was at least 1 year of outcome period. Where possible a 2-year outcome period was used. Ideally at least a 2-year baseline period would have been used, but to maximise the number of sites we could include a shorter baseline period was used if an appropriate model could be fitted to the data. The Mean Absolute Percentage Error (MAPE) and Durbin-Watson statistic of the models were checked, and the prior local level standard error adjusted if required to ensure a good historical model fit while minimising the forecasting intervals.

Aggregate (meta) analysis

Where possible, for each indicator variable a meta-analysis was conducted using the data for district general and acute hospitals. The specialist hospital sites were excluded from this analysis as it is felt the effects at these sites may not be generalisable when considering future building of general acute hospitals.

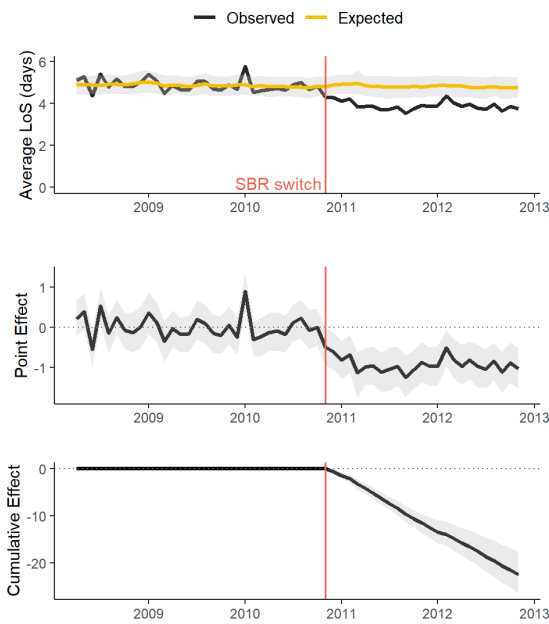
A random-effects meta-analysis was conducted using the `metafor` R package (Viechtbauer 2024), and the relative effects for each site weighted using the inverse variance, as used for meta-analysis of causal impact models by Schmitt et al. (Schmitt, Tull, and Atwater 2017).

Main results (including related notes from lookback exercise)

Interpreting the causal impact analysis

Plots for each causal impact model are shown in appendix A ([Section 12.1](#)). These plots are made up of three graphs, with the time point of the switch shown in red.

For example:



1. Expected value of the indicator in the outcome period (yellow), compared to the actual values of the study site (black).

2. The point estimate shows the difference between the expected values and actual values at each time point, so at time points where the posterior intervals do not include zero there is a significant difference.

3. Cumulative difference between the expected and actual values over the outcome period. Our measures are typically rates and percentages, which are not additive over time like count data would be, so the cumulative effect can not always be meaningfully interpreted. Also, the cumulative difference cannot always be compared across sites, due to data issues resulting in differing lengths of outcome period.

Forest plots summarise the relative effect in the outcome period, which is the proportional change following a switch to single bedrooms compared to what would be expected had no change occurred. For example, for median length of stay at Peterborough hospital the effect size is -0.19 (with a 95% credible interval of -0.22 to -0.16), indicating a 19% decrease in the median length of stay. Points to the left of the zero line on the forest plot with a credible interval that does not cross zero indicate there was a significant decrease. Since a decrease in length of stay is a positive outcome the point is coloured green. Any significant negative effect effect, such as an increase in length of stay or a decrease in positive responses to Friends and Family test, are shown in red.

Where there are results for 2 or more general acute hospitals a second forest plot shows the effects for just these general acute hospitals, alongside the mean value calculated in the meta-analysis. In this case the size of points indicates the weight that each site has been given in the analysis; those with the smallest credible intervals are given the greatest weight in calculating the mean.

Results tables give the average values over the outcome period. They give the actual value for the study site in the outcome period, the predicted value had no change in single bedrooms occurred, and the absolute effect, which is the difference between the actual and predicted value. The relative effect size is a proportional change relative to the predicted value.

For example, Peterborough hospital was predicted to have an mean length of stay in the outcome period of 4.8 days, however, following a switch to single bedrooms the actual mean length of stay was 3.9 days. This is an absolute difference between predicted and actual values of 0.9 days, which equates to a 19% reduction in mean length of stay compared to predicted values.

Utilising the relative effect - The relative effect gives us the proportional change that occurs as a result of switching to single bedrooms, so this change can be directly applied to the predicted values at any particular point

in time for a hospital with multi-bed bays to determine the impact that single bedrooms would have. For example, if length of stay at some future time point in a hospital with multi-bed bays is predicted to be 5 days, then the mean effect size of -0.08 suggests the mean length of stay at a hospital with single bedrooms would be reduced to 4.6 days.

Some of our indicators are rates, so the percentage change is a change in a rate. For example, a 10% decrease in emergency readmissions following a switch to single bedrooms means that if the predicted emergency readmission rate was 20% for a hospital with multi-bed bays, the emergency readmission rate for a single bedroom hospital would be expected to be 18%.

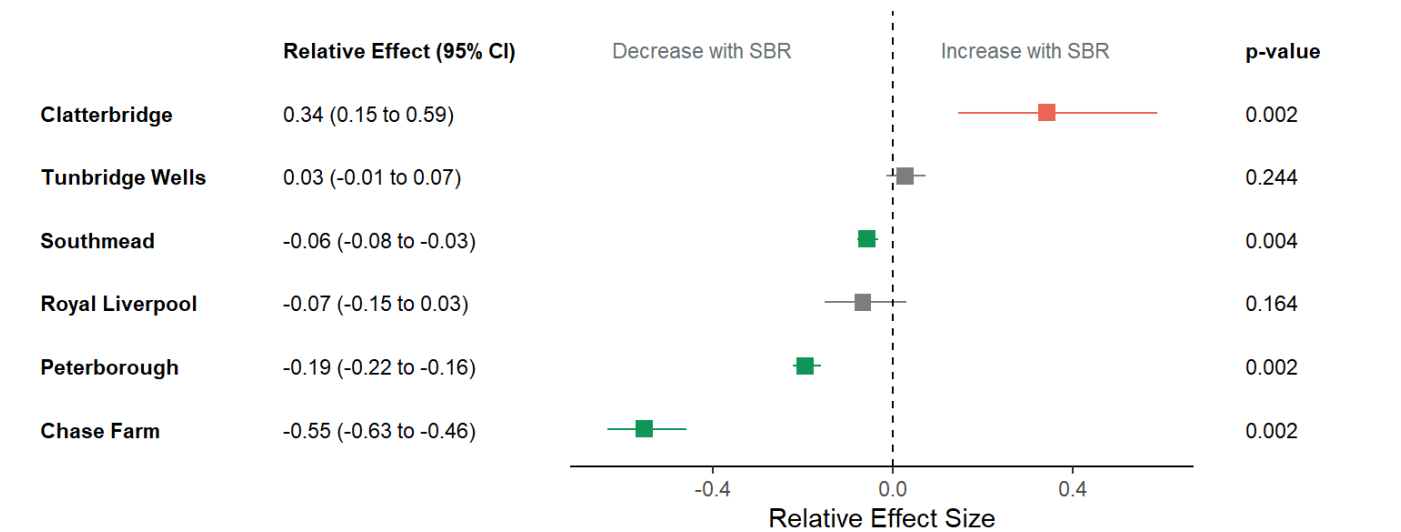
It is assumed the effects have stabilised within the 2 year outcome period so that the average relative effect in the outcome period is reflective of the true ongoing effect at any point in time. The indicators used are generally rates and percentages so any changes are not applied cumulatively. Interpretation of median waiting time is more complex; it is possible there may be cumulative impacts as a result of sustained changes in productivity affecting waiting list size.

Productivity and Efficiency

Length of stay

Mean length of stay (in days) significantly decreased at 3/6 sites (Southmead, Peterborough and Chase Farm), was unchanged at 2/6 sites (Royal Liverpool and Tunbridge Wells), but significantly increased at Clatterbridge.

Relative effect of switching to single bedrooms on mean length of stay



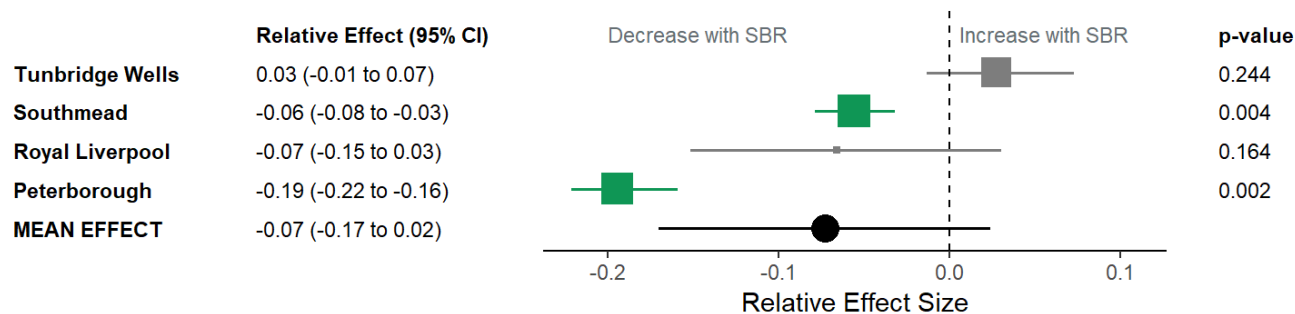
Average effects of switching to single bedrooms on mean length of stay

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Clatterbridge	10.11	7.59 (6.37 to	2.53 (1.29 to 3.75)	0.34 (0.15 to 0.59)	0.002	Sig Increase

Average effects of switching to single bedrooms on mean length of stay

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
		8.83)				
Tunbridge Wells	3.42	3.33 (3.19 to 3.46)	0.09 (-0.05 to 0.23)	0.03 (-0.01 to 0.07)	0.244	Non-sig
Southmead	4.52	4.78 (4.66 to 4.9)	-0.27 (-0.39 to -0.15)	-0.06 (-0.08 to -0.03)	0.004	Sig Decrease
Royal Liverpool	4.99	5.36 (4.84 to 5.88)	-0.37 (-0.89 to 0.15)	-0.07 (-0.15 to 0.03)	0.164	Non-sig
Peterborough	3.88	4.82 (4.62 to 4.98)	-0.94 (-1.1 to -0.74)	-0.19 (-0.22 to -0.16)	0.002	Sig Decrease
Chase Farm	2.67	6.02 (4.92 to 7.3)	-3.35 (-4.62 to -2.25)	-0.55 (-0.63 to -0.46)	0.002	Sig Decrease

Mean effect on length of stay at general acute hospitals



Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

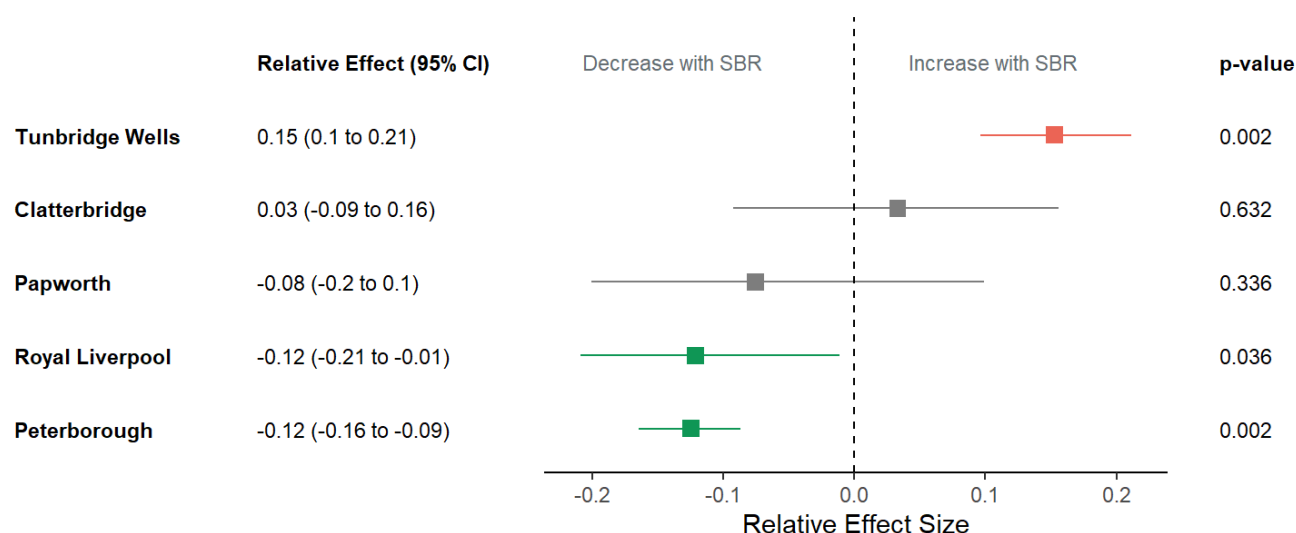
The effect on length of stay was highly variable with Chase Farm (an elective care site) having a 55% reduction in length and Clatterbridge (a cancer centre) having a 34% increase ([Section 12.1.1](#)). At the general acute hospitals the change in average length of stay varied from a 3% increase (equivalent to 0.1 days) at Tunbridge Wells to a significant 19% decrease (equivalent to 0.9 days) at Peterborough. The mean effect at general acute hospitals was a non-significant 7% reduction in length of stay.

Clatterbridge is a specialist cancer centre, so it is possible the switch to single bed rooms had a different impact on length of stay due to the nature of the activity at the site. Indeed, the average length of stay predicted for Clatterbridge based on historical trends (7.5 days) is higher than for the other sites, likely reflecting the different type of activity at this site. Average length of stay at Chase Farm was predicted to be 6 days, but following the switch was actually just over 2.5 days, such a large change may reflect a change in the type of activity conducted at the site following the switch, especially as this site conducts only elective care.

Median waiting time

The median referral to treatment waiting time (in weeks) significantly decreased at 2/5 sites (Royal Liverpool and Peterborough), was unchanged at 2/5 sites (Papworth and Clatterbridge) and increased at Tunbridge Wells.

Relative effect of switching to single bedrooms on median waiting time



Points to the left indicate a decrease in median waiting time following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the right) are indicated red. Grey indicates no significant change.

Mean effect on median waiting time at general acute hospitals

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Tunbridge Wells	5.46	4.74 (4.51 to 4.98)	0.72 (0.48 to 0.95)	0.15 (0.1 to 0.21)	0.002	Sig Increase
Clatterbridge	6.00	5.83 (5.19 to 6.61)	0.17 (-0.61 to 0.81)	0.03 (-0.09 to 0.16)	0.632	Non-sig
Papworth	8.38	9.12 (7.62 to 10.47)	-0.74 (-2.1 to 0.76)	-0.08 (-0.2 to 0.1)	0.336	Non-sig
Royal Liverpool	16.33	18.63 (16.52 to 20.64)	-2.3 (-4.3 to -0.18)	-0.12 (-0.21 to -0.01)	0.036	Sig Decrease
Peterborough	4.33	4.95 (4.75 to 5.18)	-0.62 (-0.85 to -0.41)	-0.12 (-0.16 to -0.09)	0.002	Sig Decrease

Mean effect on median waiting time at general acute hospitals



Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

The effect on median waiting time varied from a 12% reduction to a 15% increase. Both Royal Liverpool and Peterborough had a 12% reduction in median waiting time, which equated to a decrease from 18.6 wks to 16.3 wks at Royal Liverpool and from 5 wks to 4.3 wks at Peterborough. The mean effect at general acute hospitals was a non-significant 3% reduction in median waiting time.

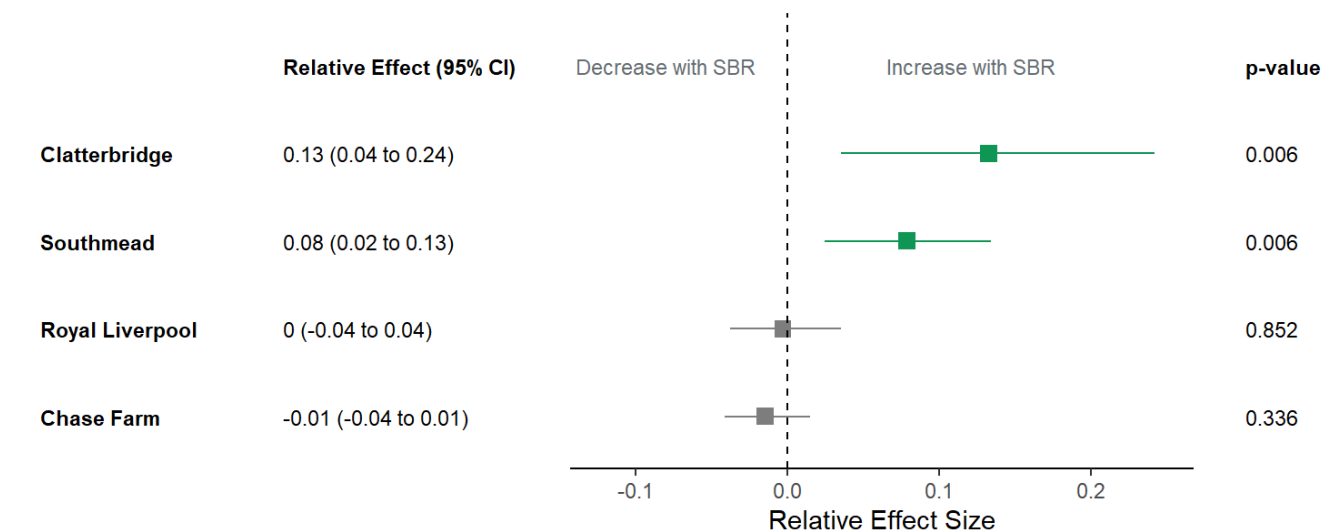
At Peterborough median waiting time dropped and was then relatively stable for the 2-year outcome period suggesting there was a fixed effect and that median waiting time would be unlikely to drop further over time as a result of the switch to single bedrooms ([Section 12.1.2](#)).

Median waiting time significantly increased at Tunbridge Wells, but this effect appeared to only be temporary during the first year post-switch as during the second year of the outcome period the waiting times fall back in line with the expected values. The switch to single bedrooms at Tunbridge Wells started in Jan 2011, which is the date we have used in our analysis, but occurred in stages between Jan and Sept 2011, so the differences seen in the first year could be due to disruption of services while the hospital was in a transition period.

Bed occupancy

Bed occupancy is measured quarterly and is only available at provider level. The percentage of available beds occupied significantly increased at 2/4 sites (Clatterbridge and Southmead) and was unchanged at 2/4 (Royal Liverpool and Chase Farm). An increase in bed occupancy is generally considered a positive result as it indicates a hospital is able to make greater use of its beds, as a result of not needing to leave beds empty to prevent spread of infection or maintain single sex wards. However, if bed occupancy is too high it can be considered negative as it indicates a potential lack of capacity and this can result in an increase in Emergency department waiting times as there are not sufficient beds to admit patients when required. We have not looked at Emergency department waiting times. However, Chase Farm only does elective care and Clatterbridge is a cancer centre, and so neither of these sites have Emergency departments.

Relative effect of switching to single bedrooms on bed occupancy



Points to the right indicate an increase in bed occupancy following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the left) are indicated red. Grey indicates no significant change.

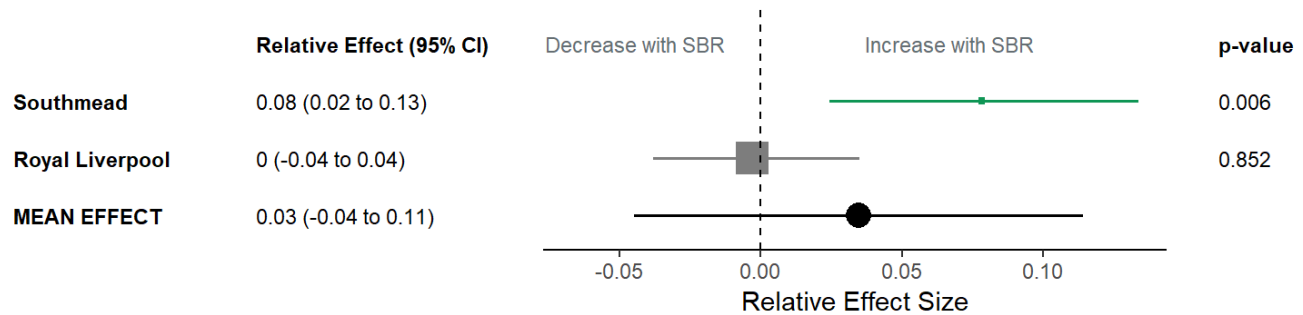
Average effects of switching to single bedrooms on % of bed occupancy

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Clatterbridge	74.33	65.77 (59.85 to 71.82)	8.56 (2.52 to 14.48)	0.13 (0.04 to 0.24)	0.006	Sig Increase
Southmead	93.19	86.47 (82.19 to 90.75)	6.73 (2.21 to 11)	0.08 (0.02 to 0.13)	0.006	Sig Increase

Average effects of switching to single bedrooms on % of bed occupancy

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
		90.98)		0.13)		
Royal Liverpool	95.19	95.45 (91.97 to 98.96)	-0.27 (-3.77 to 3.22)	0 (-0.04 to 0.04)	0.852	Non-sig
Chase Farm	85.25	86.51 (84.02 to 88.92)	-1.26 (-3.67 to 1.23)	-0.01 (-0.04 to 0.01)	0.336	Non-sig

Mean effect on bed occupancy at general acute hospitals



Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

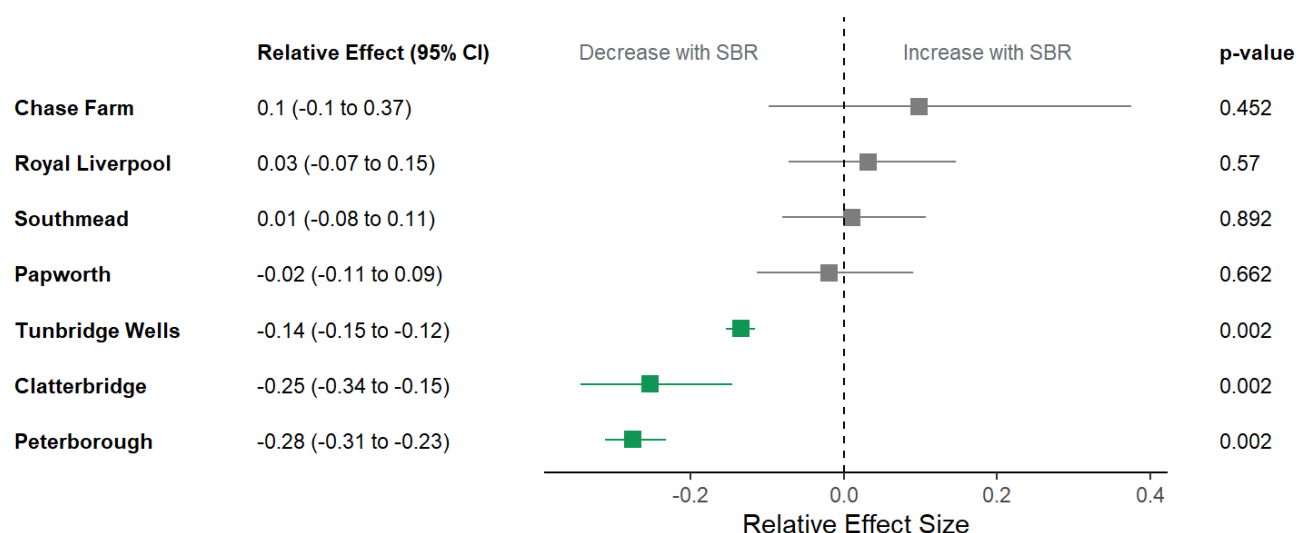
Bed occupancy was either unchanged or increased by up to 13%. The mean effect at the two general acute hospitals was a non-significant 3% increase in bed occupancy.

At Southmead, bed occupancy increased 8%; it was predicted to be on average 86% in the outcome period but an increase to 93% bed occupancy was seen ([Section 12.1.3](#)). It is perhaps unsurprising there was no significant increase in bed occupancy at the Royal Liverpool as based on historic trends bed occupancy was already predicted to be over 95%, leaving little scope to increase this.

Emergency readmissions

The percentage of admissions that are emergency readmissions significantly decreased at 3/7 sites (Tunbridge Wells, Peterborough and Clatterbridge) and was unchanged at the remaining 4/7 sites.

Relative effect of switching to single bedrooms on emergency readmissions



Points to the left indicate a decrease in emergency readmissions following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the right) are red. Grey indicates no significant change.

Average effects of switching to single bedrooms on % of emergency readmissions

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Chase Farm	7.68	7.04 (5.59 to 8.52)	0.64 (-0.83 to 2.09)	0.1 (-0.1 to 0.37)	0.452	Non-sig
Royal Liverpool	50.97	49.54 (44.47 to 54.95)	1.43 (-3.98 to 6.5)	0.03 (-0.07 to 0.15)	0.57	Non-sig
Southmead	26.37	26.16 (23.81 to 28.68)	0.21 (-2.3 to 2.56)	0.01 (-0.08 to 0.11)	0.892	Non-sig
Papworth	12.00	12.27 (11.01 to 13.55)	-0.26 (-1.55 to 1)	-0.02 (-0.11 to 0.09)	0.662	Non-sig
Tunbridge Wells	26.05	30.11 (29.47 to 30.78)	-4.06 (-4.73 to -3.42)	-0.14 (-0.15 to -0.12)	0.002	Sig Decrease
Clatterbridge	25.04	33.65 (29.31 to 38.17)	-8.61 (-13.13 to -4.27)	-0.25 (-0.34 to -0.15)	0.002	Sig Decrease
Peterborough	24.85	34.32 (32.39 to 36.11)	-9.47 (-11.26 to -7.54)	-0.28 (-0.31 to -0.23)	0.002	Sig Decrease

Mean effect on emergency readmission rate at general acute hospitals



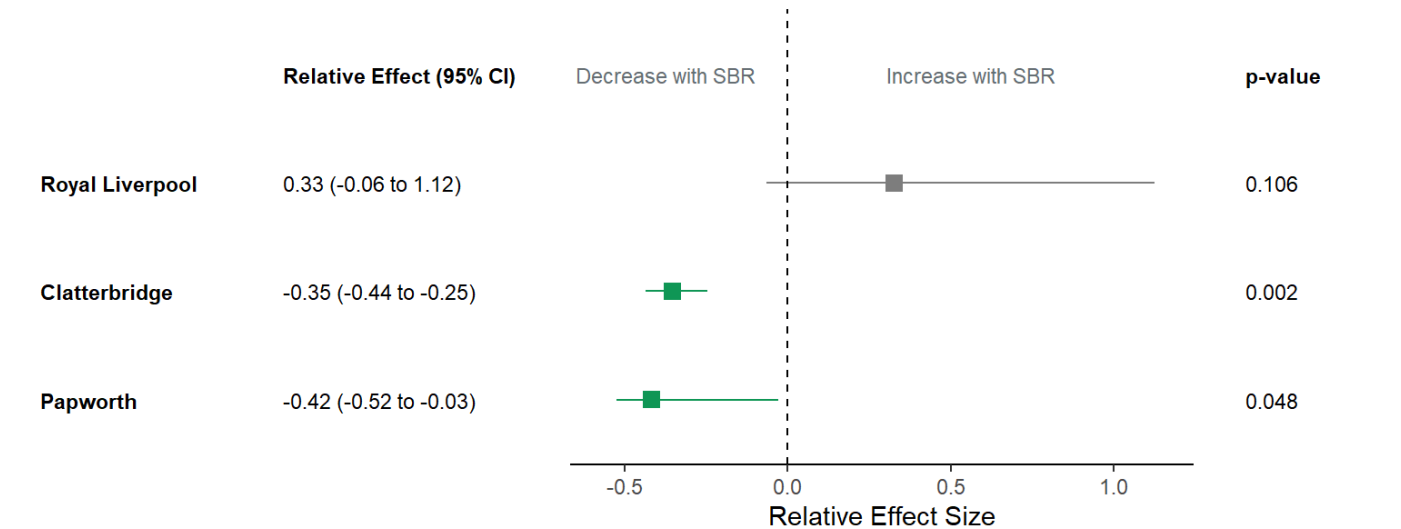
Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

The emergency readmission rate was either unchanged or decreased by up to 28%. The mean effect at general acute hospitals was a non-significant 10% reduction in emergency readmissions. At Peterborough the percentage of emergency readmissions was predicted to be 34% but dropped to 25% following a switch to single bedrooms, while at Tunbridge Wells it was predicted to be 30% and dropped to 26% ([Section 12.1.4](#)).

Cleaning costs

The cleaning costs (in £) per square metre of occupied floor space significantly decreased at 2/3 sites (Clatterbridge and Papworth) but were unchanged at 1/3 sites (Royal Liverpool).

Relative effect of switching to single bedrooms on cleaning costs



Points to the left indicate a decrease in cleaning costs following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the right) are red. Grey indicates no significant change.

Average effects of switching to single bedrooms on cleaning costs (£) per sqm

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Royal Liverpool	56.44	44.18 (26.56 to 60.3)	12.26 (-3.86 to 29.88)	0.33 (-0.06 to 1.12)	0.106	Non-sig
Clatterbridge	32.75	50.91 (43.48 to 58.06)	-18.16 (-25.31 to -10.74)	-0.35 (-0.44 to -0.25)	0.002	Sig Decrease
Papworth	32.78	53.98 (33.54 to 68.63)	-21.2 (-35.85 to -0.76)	-0.42 (-0.52 to -0.03)	0.048	Sig Decrease

Actual, Predicted and Absolute difference values expressed as £ per sqm

Cleaning costs decreased by 35% and 42% at Clatterbridge and Papworth, respectively ([Section 12.1.5](#)).

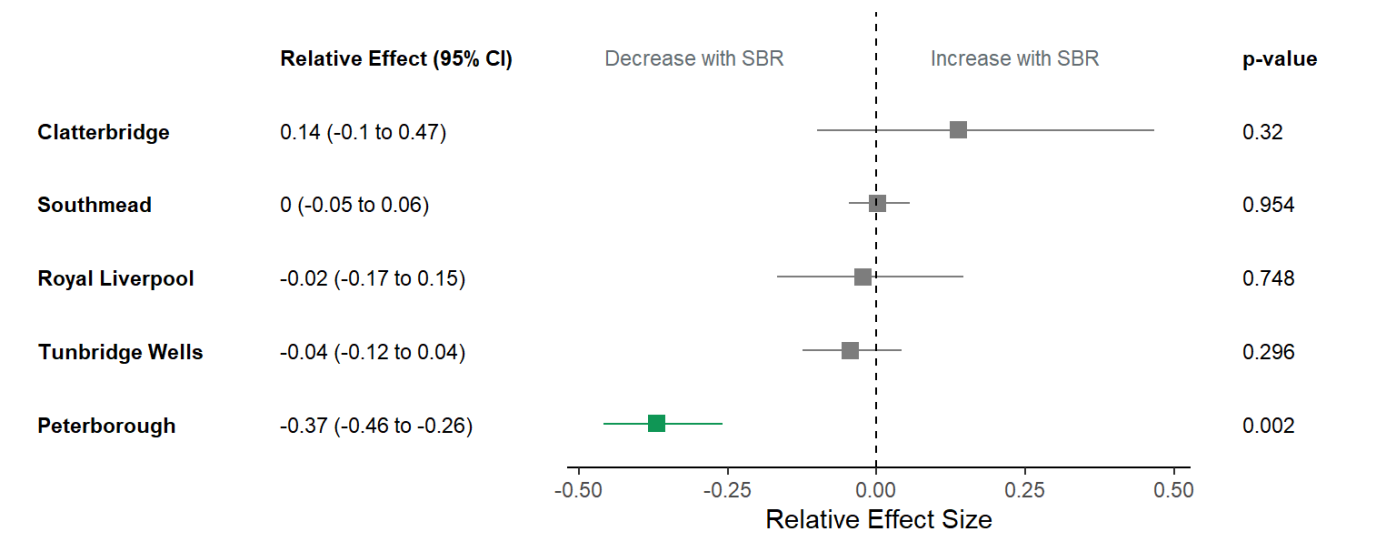
Cleaning costs are only recorded annually, so there is limited data with only 1 or 2 data points in the outcome period, and the first data point in the outcome period may not reflect the true changes in cost as some of the costs will have occurred prior to the switch, so these results should be interpreted cautiously.

Health and Safety

Deaths in hospital

The rate of in-hospital deaths (number per 1,000 discharges) significantly decreased at 1/5 sites (Peterborough) and was unchanged at 4/5 sites (Royal Liverpool, Tunbridge Wells, Clatterbridge and Southmead).

Relative effect of switching to single bedrooms on in-hospital deaths

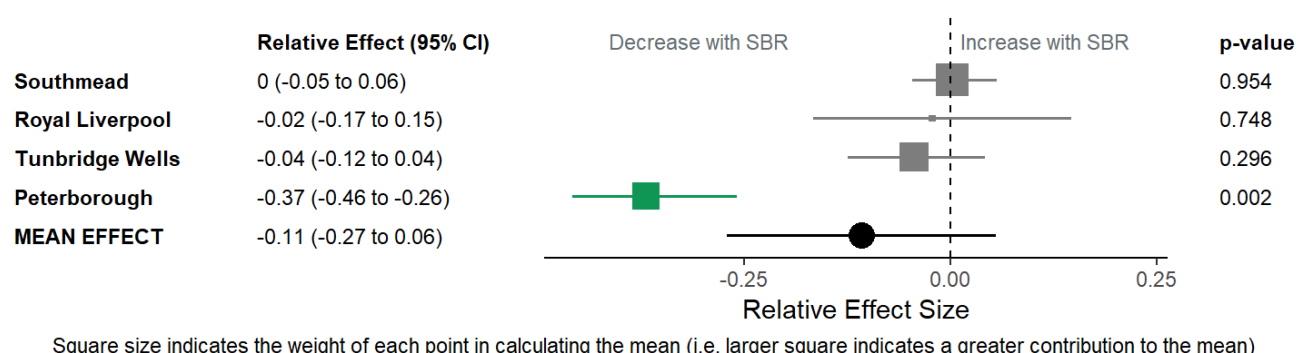


Points to the left indicate a decrease in hospital deaths following a switch to single bed rooms, which is considered a positive effect. Significant positive effects green, while significant negative effects (on the right) are red. Grey indicates no significant change.

Average effects of switching to single bedrooms on in-hospital deaths/1,000 discharges

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Clatterbridge	85.80	76.41 (58.49 to 95.41)	9.39 (-9.61 to 27.31)	0.14 (-0.1 to 0.47)	0.32	Non-sig
Southmead	25.87	25.81 (24.5 to 27.14)	0.05 (-1.27 to 1.37)	0 (-0.05 to 0.06)	0.954	Non-sig
Royal Liverpool	37.37	38.46 (32.6 to 44.87)	-1.09 (-7.5 to 4.78)	-0.02 (-0.17 to 0.15)	0.748	Non-sig
Tunbridge Wells	18.06	18.9 (17.34 to 20.63)	-0.84 (-2.56 to 0.73)	-0.04 (-0.12 to 0.04)	0.296	Non-sig
Peterborough	23.96	38.29 (32.35 to 44.27)	-14.33 (-20.3 to -8.39)	-0.37 (-0.46 to -0.26)	0.002	Sig Decrease

Mean effect on in-hospital death rate at general acute hospitals

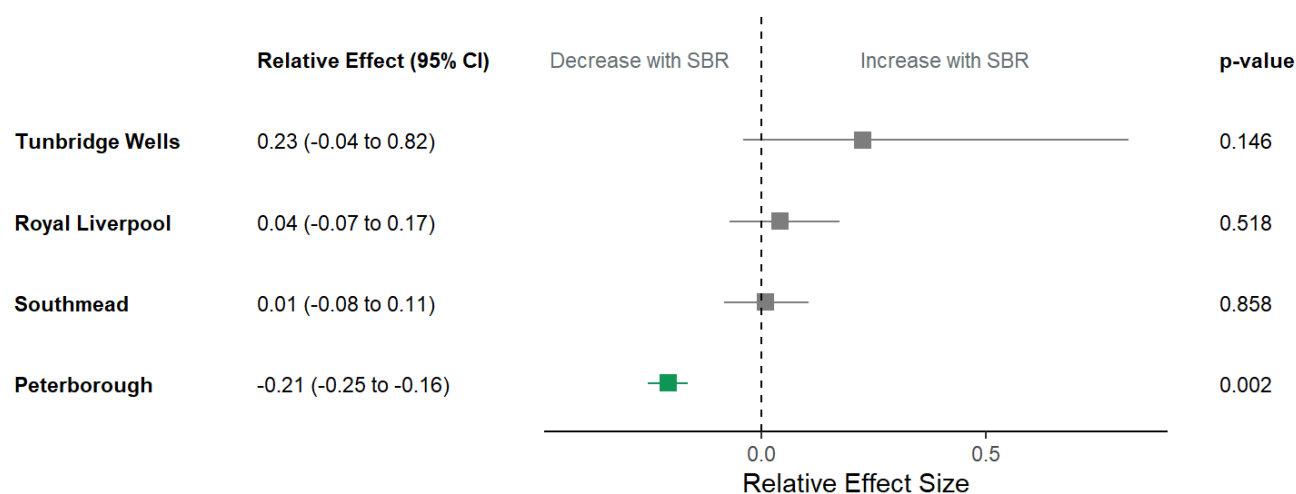


In-hospital deaths were generally unchanged but decreased by 37% at Peterborough ([Section 12.1.6](#)). The 37% decrease at Peterborough reflects a decrease from a predicted 38 deaths per 1,000 discharges to 24 deaths. The mean effect at general acute hospitals was a non-significant decrease of 11% in in-hospital death.

Falls and injuries in hospital

The rate of in-hospital falls and injuries (% of spells) significantly decreased at 1/4 sites (Peterborough), and was unchanged at 3/4 sites (Royal Liverpool, Tunbridge Wells and Southmead).

Relative effect of switching to single bedrooms on in-hospital falls, fractures and injuries



Points to the left indicate a decrease in falls and fractures following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the right) are red. Grey indicates no significant change.

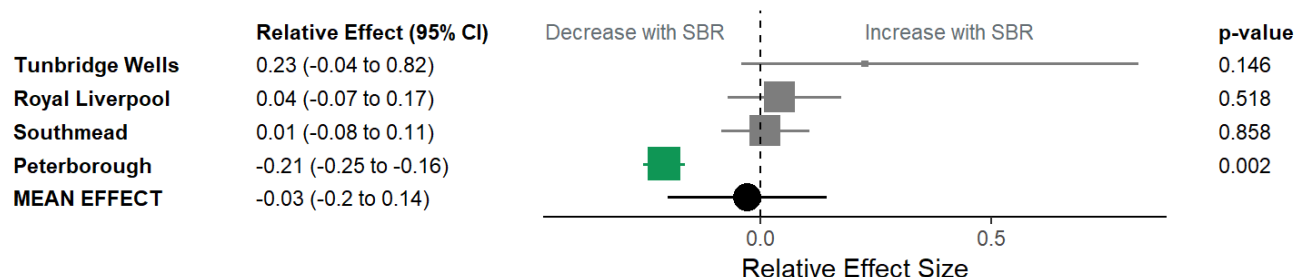
Average effects of switching to single bedrooms on % of spells with in-hospital fall or injury

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Tunbridge Wells	1.05	0.88 (0.58 to 1.09)	0.17 (-0.04 to 0.47)	0.23 (-0.04 to 0.82)	0.146	Non-sig
Royal Liverpool	3.43	3.31 (2.92 to 3.69)	0.13 (-0.26 to 0.51)	0.04 (-0.07 to 0.17)	0.518	Non-sig

Average effects of switching to single bedrooms on % of spells with in-hospital fall or injury

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Southmead	1.60	1.58 (1.44 to 1.74)	0.01 (-0.15 to 0.15)	0.01 (-0.08 to 0.11)	0.858	Non-sig
Peterborough	2.54	3.21 (3.04 to 3.41)	-0.67 (-0.87 to -0.5)	-0.21 (-0.25 to -0.16)	0.002	Sig Decrease

Mean effect on rate of falls, fractures and injuries while in hospital at general acute hospitals



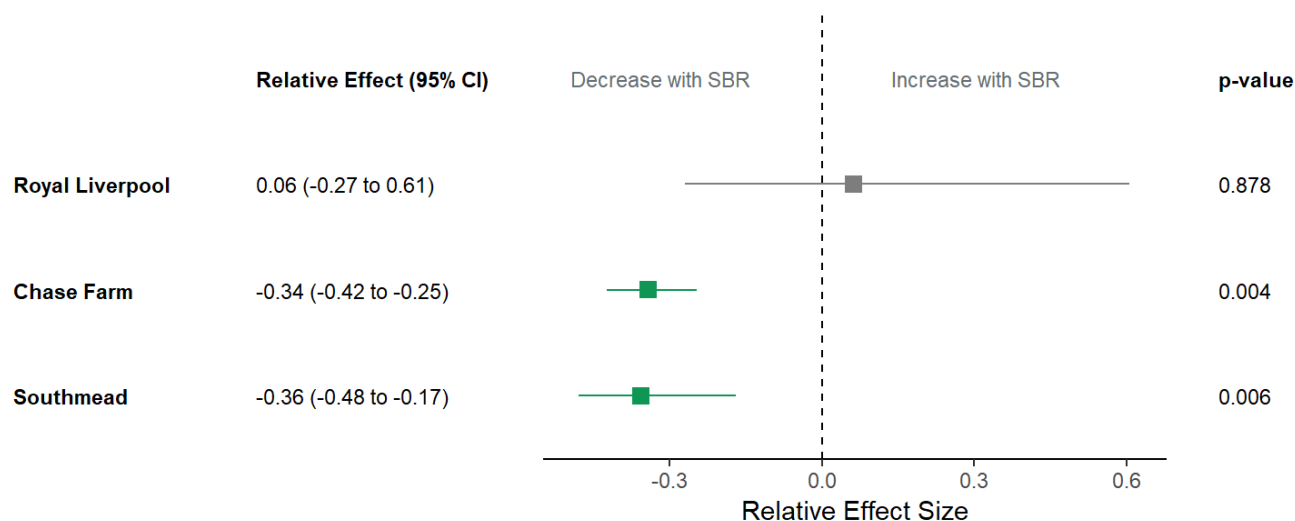
Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

The in-hospital falls and injuries rate was generally unchanged, but decreased by 21% at Peterborough ([Section 12.1.7](#)). The 21% decrease at Peterborough reflects a decrease from a predicted 3.2% of hospital spells having a fall or injury to 2.5%. The mean effect at general acute hospitals was a non-significant 3% reduction in the rate of in-hospital falls, injuries and fractures.

Healthcare acquired c.difficile infections

The rate of healthcare acquired C. difficile infections (cases per 10,000 bed days) decreased at 2/3 sites (Chase Farm and Southmead) but was unchanged at Royal Liverpool.

Relative effect of switching to single bedrooms on rate of healthcare acquired C.Difficile infections

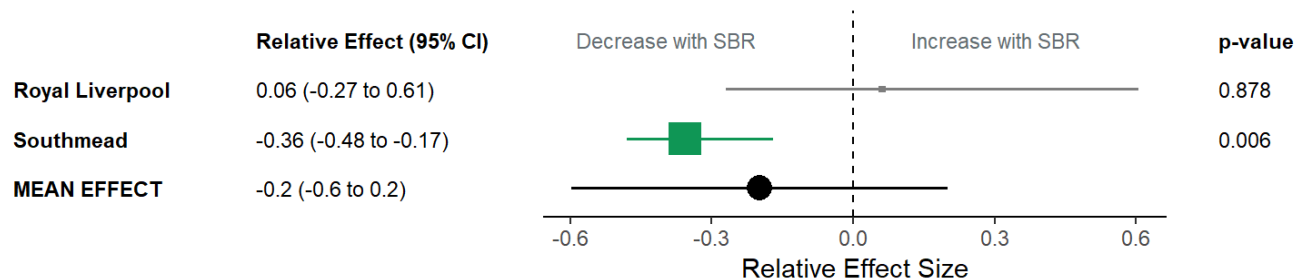


Points to the left indicate a decrease in healthcare acquired C. Difficile infections following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green. Grey indicates no significant change.

Average effects of switching to single bedrooms on cases of healthcare acquired C.Difficile infections /10,000 bed days

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Royal Liverpool	2.45	2.39 (1.52 to 3.36)	0.06 (-0.91 to 0.92)	0.06 (-0.27 to 0.61)	0.878	Non-sig
Chase Farm	2.65	4.04 (3.52 to 4.6)	-1.39 (-1.95 to -0.87)	-0.34 (-0.42 to -0.25)	0.004	Sig Decrease
Southmead	1.33	2.09 (1.6 to 2.55)	-0.76 (-1.22 to -0.27)	-0.36 (-0.48 to -0.17)	0.006	Sig Decrease

Mean effect on the rate healthcare acquired C.Difficile infections at general acute hospitals



Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

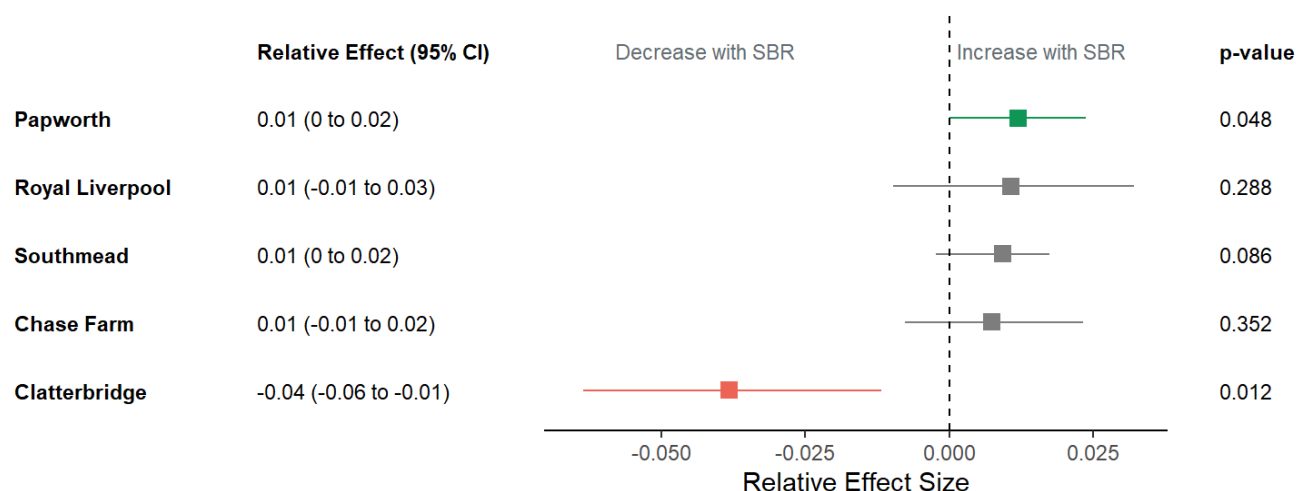
The rate of healthcare acquired C. difficile was either unchanged or decreased by up to 36% ([Section 12.1.8](#)). The 36% decrease in C.diff infections at Southmead reflects a decrease from a predicted 2.1 cases per 10,000 beddays to 1.3 cases. The mean effect at general acute hospitals was a non-significant 20% reduction in the rate of C. difficile infections.

Patient and Staff Experience

Patient experience - Friends and Family Test

Patient experience was determined using the results of the Friends and Family test. The percentage of positive responses significantly decreased at 1/5 sites (Clatterbridge), increased at 1/5 sites (Papworth) and was unchanged at the other 3/5 sites (Royal Liverpool, Southmead and Chase Farm).

Relative effect of switching to single bedrooms on positive scores on the Friends and Family test

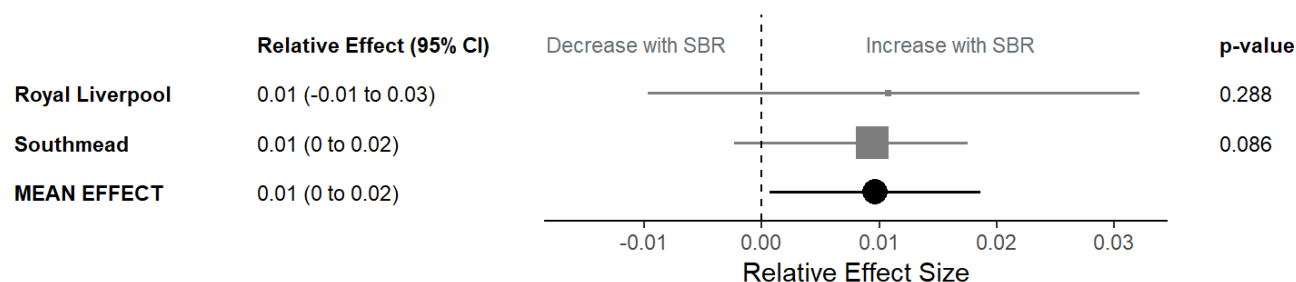


Points to the right indicate an increase in positive scores on the Friends and Family patient experience test following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the left) are red. Grey indicates no significant change.

Average effects of switching to single bedrooms on % of positive responses on the Friends and Family test

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Papworth	97.92	96.76 (95.65 to 97.91)	1.16 (0.01 to 2.27)	0.01 (0 to 0.02)	0.048	Sig Increase
Royal Liverpool	91.41	90.43 (88.56 to 92.3)	0.98 (-0.9 to 2.85)	0.01 (-0.01 to 0.03)	0.288	Non-sig
Southmead	95.06	94.18 (93.43 to 95.28)	0.88 (-0.22 to 1.63)	0.01 (0 to 0.02)	0.086	Non-sig
Chase Farm	94.82	94.09 (92.67 to 95.56)	0.73 (-0.73 to 2.16)	0.01 (-0.01 to 0.02)	0.352	Non-sig
Clatterbridge	93.87	97.61 (94.99 to 100.24)	-3.74 (-6.37 to -1.12)	-0.04 (-0.06 to -0.01)	0.012	Sig Decrease

Mean effect on the % of positive responses to the Friends and Family test at general acute hospitals



Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

The effect on the percentage of positive responses to the Friend and Family Test was mixed; the percentage of positive responses increased by 1% at Papworth, but decreased by 4% at Clatterbridge ([Section 12.1.9](#)). It should be noted that based on historic trends Clatterbridge was predicted to have over 97% of positive responses, so

although there was a small decrease the percentage of positive responses remained at almost 94%, which is broadly similar to that seen at the other study sites.

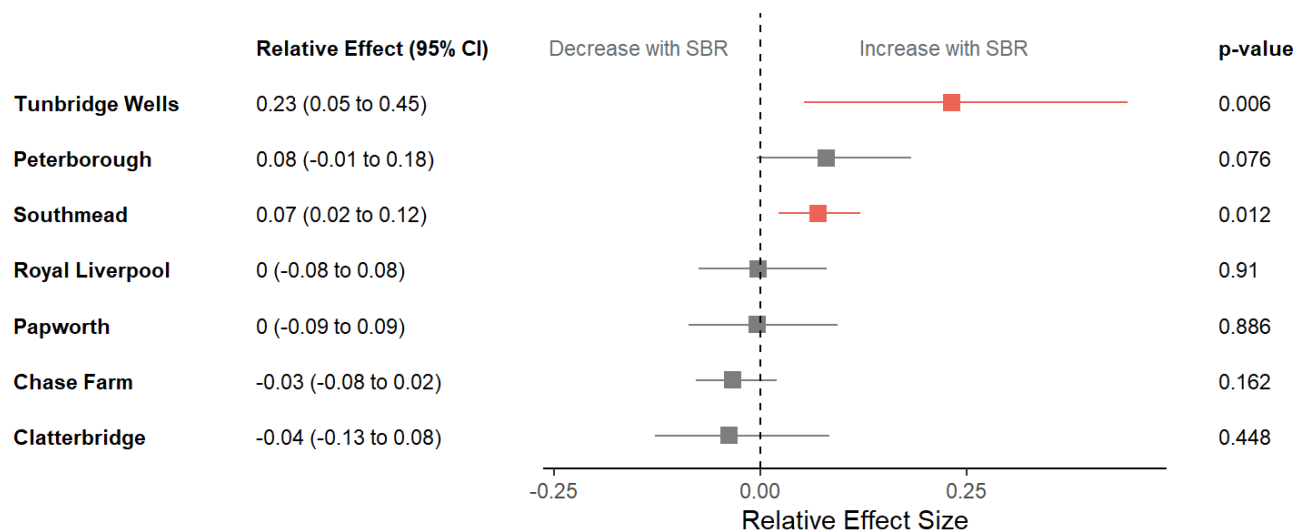
For the general acute hospitals the mean effect was a statistically significant 1% increase in positive responses to the Friends and Family test.

It should be noted that due to the COVID-19 pandemic there is missing friends and family test scores data from March 2020 until late 2020/early 2021 for all sites. Therefore, both Papworth and Clatterbridge have missing data in their outcome periods.

Staff sickness

The percentage of available days taken off sick by staff significantly increased at 2/7 sites (Southmead and Tunbridge Wells) and was unchanged at the remaining 5 sites.

Relative effect of switching to single bedrooms on staff sickness rate



Points to the left indicate a decrease in staff sickness following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the right) are red. Grey indicates no significant change.

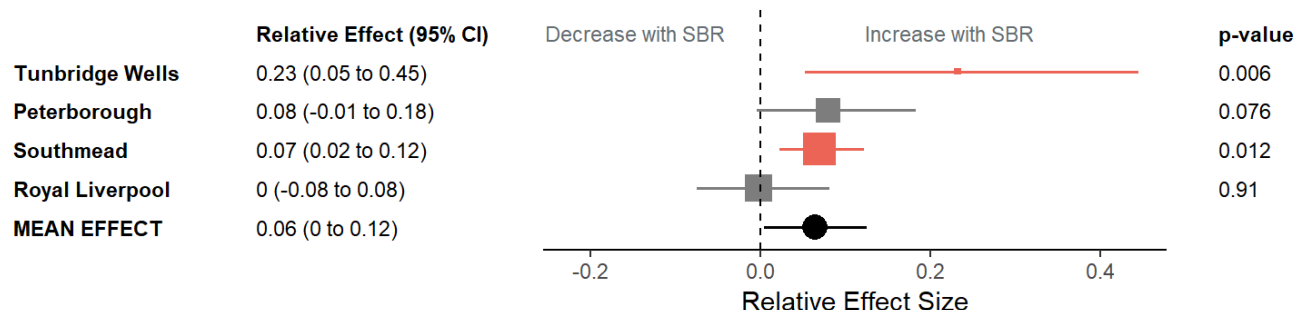
Average effects of switching to single bedrooms on staff sickness rate (% of available days)

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Tunbridge Wells	3.63	2.96 (2.51 to 3.45)	0.66 (0.18 to 1.12)	0.23 (0.05 to 0.45)	0.006	Sig Increase
Peterborough	3.57	3.32 (3.02 to 3.59)	0.26 (-0.02 to 0.55)	0.08 (-0.01 to 0.18)	0.076	Non-sig
Southmead	4.64	4.34 (4.14 to 4.54)	0.3 (0.1 to 0.5)	0.07 (0.02 to 0.12)	0.012	Sig Increase
Royal Liverpool	6.34	6.36 (5.87 to 6.86)	-0.03 (-0.52 to 0.47)	0 (-0.08 to 0.08)	0.91	Non-sig
Papworth	3.77	3.79 (3.45 to 4.13)	-0.02 (-0.36 to 0.32)	0 (-0.09 to 0.09)	0.886	Non-sig

Average effects of switching to single bedrooms on staff sickness rate (% of available days)

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Chase Farm	3.64	3.77 (3.57 to 3.95)	-0.13 (-0.31 to 0.07)	-0.03 (-0.08 to 0.02)	0.162	Non-sig
Clatterbridge	4.91	5.11 (4.53 to 5.63)	-0.21 (-0.72 to 0.38)	-0.04 (-0.13 to 0.08)	0.448	Non-sig

Mean effect on the rate of staff sickness at general acute hospitals



Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

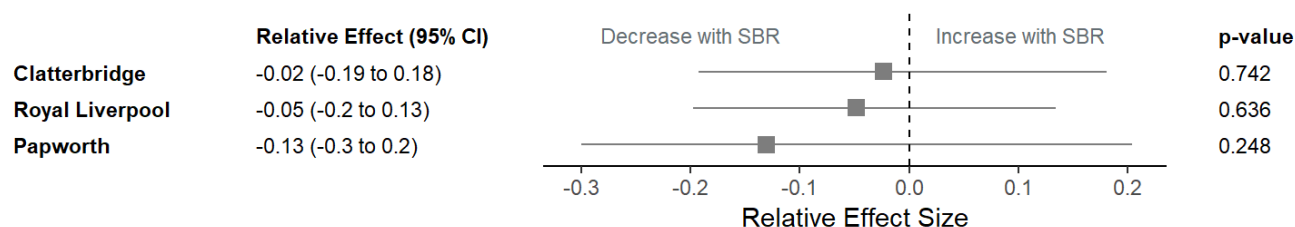
Staff sickness rates generally remained unchanged, but there was a 7% increase at Southmead and 23% increase at Tunbridge Wells ([Section 12.1.10](#)). The mean effect at general acute hospitals was a significant increase of 6% in the staff sickness rate. Although staff sickness did remain unchanged at the three specialist sites.

While there was a significant increase in staff sickness at Southmead (average of 4.6% of available days taken as sick days compared to a predicted 4.3%) this seems to have been a temporary effect. The sickness rate is increased in the first year after the switch to single bedrooms and during this period there is a significantly larger winter peak. In the second year of the outcome period staff sickness rates fell back in line with the expected rates. A similar, albeit non-significant, effect is seen at Peterborough with a larger winter spike in the first year of the outcome period before sickness levels return to expected levels.

Staff turnover

Staff turnover measured as percentage of staff leaving was not significantly changed at any of the three sites where data was available ([Section 12.1.11](#)).

Relative effect of switching to single bedrooms on staff turnover



Grey indicates no significant change in staff turnover following a switch to single bed rooms.

Average effects of switching to single bedrooms on rate of staff turnover (% leaving)

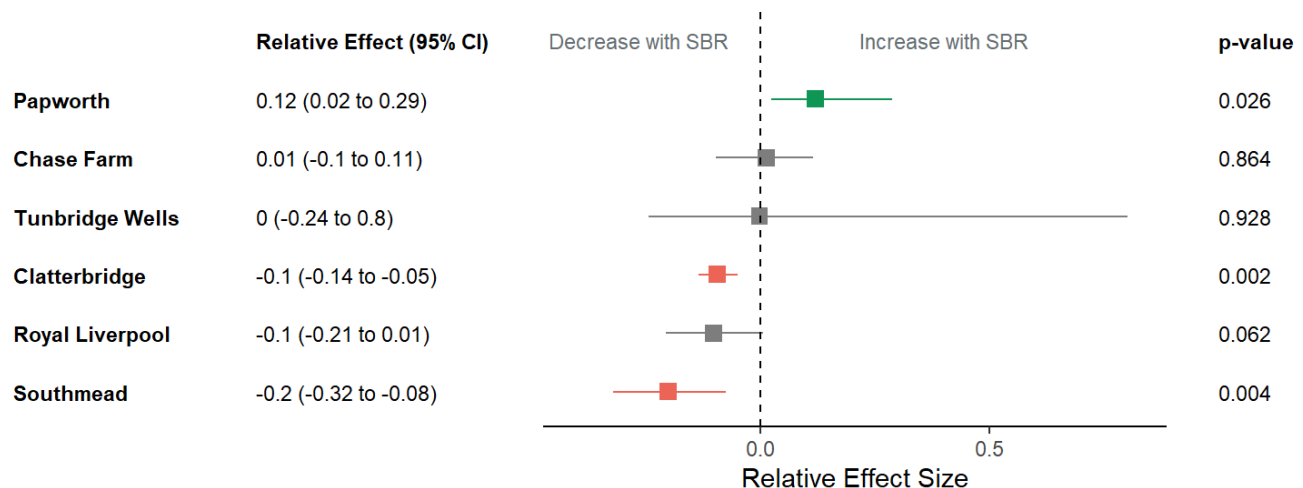
Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Clatterbridge	12.13	12.55 (10.27 to 15.03)	-0.42 (-2.9 to 1.86)	-0.02 (-0.19 to 0.18)	0.742	Non-sig
Royal Liverpool	9.20	9.76 (8.11 to 11.47)	-0.56 (-2.27 to 1.09)	-0.05 (-0.2 to 0.13)	0.636	Non-sig
Papworth	12.64	14.8 (10.5 to 18.05)	-2.16 (-5.41 to 2.14)	-0.13 (-0.3 to 0.2)	0.248	Non-sig

Staff survey

To assess staff experience the NHS staff survey was used and we analysed the percentage of staff that Agreed or Strongly Agreed with the statement 'I would recommend my trust as a place to work'.

The percentage of staff that would recommend their trust as a place to work significantly increased at 1/6 sites (Papworth) and was unchanged at 3/6 (Chase Farm, Royal Liverpool and Tunbridge Wells). At 2/6 sites (Southmead and Clatterbridge) there was a significant decrease in the percentage that would recommend their trust as a place to work.

Relative effect of switching to single bedrooms on % of NHS staff that recommend their workplace



Points to the right indicate an increase in positive responses to the NHS staff survey following a switch to single bed rooms, which is considered a positive effect. Significant positive effects are green, while significant negative effects (on the left) are red. Grey indicates no significant change.

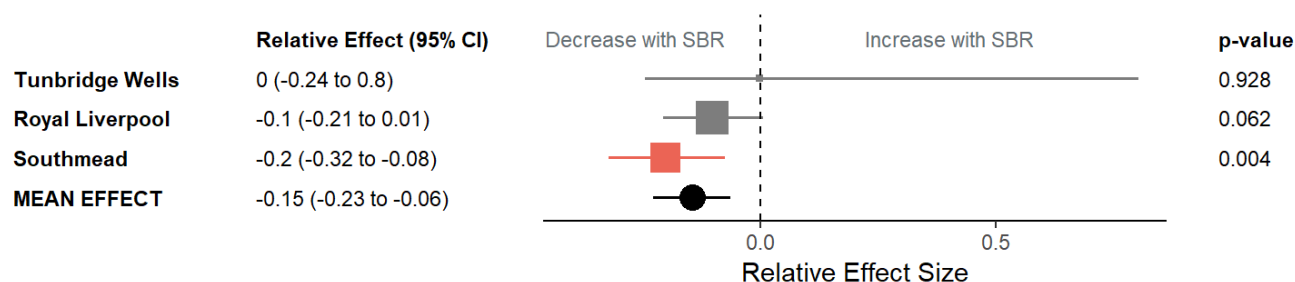
Average effects of switching to single bedrooms on % of NHS staff who would recommend their workplace

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Papworth	67.44	60.46 (52.39 to 65.87)	6.98 (1.56 to 15.05)	0.12 (0.02 to 0.29)	0.026	Sig Increase

Average effects of switching to single bedrooms on % of NHS staff who would recommend their workplace

Site	Actual Effect	Predicted Effect (95% CI)	Absolute Effect (95% CI)	Relative Effect (95% CI)	p-value	Significance
Chase Farm	62.47	61.94 (56.04 to 69.22)	0.54 (-6.75 to 6.44)	0.01 (-0.1 to 0.11)	0.864	Non-sig
Tunbridge Wells	60.28	59.12 (33.38 to 78.25)	1.16 (-17.97 to 26.9)	0 (-0.24 to 0.8)	0.928	Non-sig
Clatterbridge	64.33	71.18 (67.76 to 74.42)	-6.85 (-10.09 to -3.43)	-0.1 (-0.14 to -0.05)	0.002	Sig Decrease
Royal Liverpool	50.32	56.31 (50.04 to 63.41)	-5.99 (-13.09 to 0.28)	-0.1 (-0.21 to 0.01)	0.062	Non-sig
Southmead	45.07	56.83 (48.74 to 66.58)	-11.77 (-21.51 to -3.68)	-0.2 (-0.32 to -0.08)	0.004	Sig Decrease

Mean effect on percentage of staff that would recommend their trust as a place to work at general acute hospitals



Square size indicates the weight of each point in calculating the mean (i.e. larger square indicates a greater contribution to the mean)

The effect on the percentage of staff that would recommend their workplace varied from a 20% decrease to a 12% increase ([Section 12.1.12](#)). At Southmead the 20% reduction in the percentage of staff recommending their workplace equates to a reduction from a predicted percentage of 57% recommending their workplace to 45% following the switch to single bedrooms. The mean effect at general acute hospitals was a significant reduction of 15% in the percentage of staff who would recommend their workplace.

At Southmead the decrease appears to be temporary, occurring in the first year post-switch to single bed rooms with positive responses increasing back to the expected level in the second year of the outcome period.

The staff survey is an annual measure so there is limited data, with only 1 or 2 data points in the outcome period. The staff survey data is collected over a period of time in the autumn each year, but for analysis purposes we have set the date as the last day of the year in which the data was collected. Both Chase Farm and Royal Liverpool switched to single rooms during the autumn, so the data for the first data point in the outcome period would actually have been collected during the transition period which may have had an impact on the results.

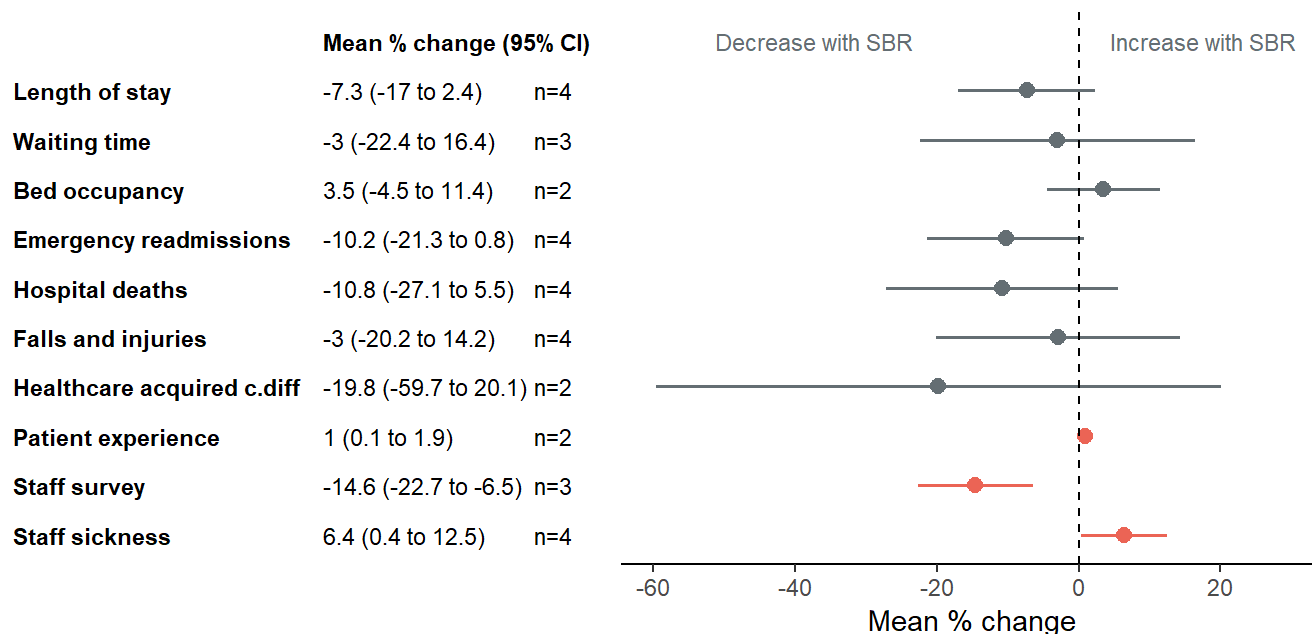
Summary of results

	Royal Liverpool	Clatterbridge	Papworth	Chase Farm	Southmead	Tunbridge Wells	Peterborough
Productivity & Efficiency							
Length of stay	-	↑		↓	↓	-	↓
Waiting time	↓	-	-			↑	↓
Bed occupancy	-	↑		-	↑		
Emergency readmissions	-	↓	-	-	-	↓	↓
Cleaning costs	-	↓	↓				
Health & Safety							
Hospital deaths	-	-			-	-	↓
Falls and injuries	-				-	-	↓
Healthcare acquired C.diff	-			↓	↓		
Patient & Staff experience							
Patient experience	-	↓	↑	-	-		
Staff sickness	-	-	-	-	↑	↑	-
Staff turnover	-	-	-				
Staff survey	-	↓	↑	-	↓	-	

↑ indicates a significant increase post-switch to single rooms, while ↓ indicates a significant decrease. Where a result is considered positive, e.g. decreased length of stay or increased patient experience score, it is coloured green, while a negative result is shown in red.

'-' indicates a non-significant change, which is shown in yellow. Grey shading indicates where there was insufficient data to analyse.

Summary of mean % changes for general acute hospitals



Grey indicates no significant effect, while red indicates a significant negative effect

The impact of switching to single bedrooms appears to vary depending on the hospital site. The switch at Peterborough has generally resulted in positive impacts, while the switch at Clatterbridge has been a little more negative. It should be noted that in some cases where negative outcomes are seen these are transient changes or there are other factors that may be influencing the results. For example, Clatterbridge switched to single bed rooms during the high of the COVID-19 pandemic and indeed its opening was brought forward as a result of the pandemic, which may have impacted outcomes.

Generally, following a switch to single bedrooms the productivity and efficiency measures are showing trends towards positive impacts. While overall at general acute hospitals there are no significant changes, the data trends towards a decrease in mean length of stay and rate of emergency readmissions and increase in bed occupancy. For median waiting time there is also no significant change overall, but results will be complicated by the dynamic nature of waiting lists, where list size not only depends on productivity but also the number of referrals, which may change with a new hospital site. Data for cleaning costs is very limited and the results are mixed.

Switching to single bedrooms appears to have a positive or neutral effect on Health and Safety measures. It could be considered that no change in health and safety measures is a positive effect, as patient safety does not appear to be negatively impacted, which has been raised as a concern due to patients being potentially less visible to staff when in single bedrooms. While our data on hospital acquired infections is very limited, there is a trend towards a decrease in C. difficile infections. At some hospital sites, such as Papworth, the rates of C.difficile rates were so low it was not possible to fit a suitable causal impact model; given the hospitals have already have very low rates it could be concluded that it would be unlikely that the rates could be further reduced by a meaningful amount.

In terms of patient and staff experiences the switch to single bedrooms has had a generally negative effect for staff. For general acute hospitals there was a significant reduction of 15% in the percentage of staff who would recommend their trust as a place to work and a significant 6% increase in staff sickness. Some of these negative effects are seen primarily in the first year post-switch before returning to expected levels during the second year. This initial negative impact may be associated with uncertainty, initial concerns over or difficulties adapting to the new design, or it may be that after time changes in policies and procedures have been introduced to mitigate the negative impacts on staff. Some staff may be negatively impacted simply by a switch to a new site, regardless of whether it has single bed rooms, due to factors such as longer or more complex travel to the new hospital site. There was an increase in staff sickness, suggesting that staff are not better protected from contracting infections from patients in single room hospitals, but that potentially staff could be having increased sickness potentially related to musculoskeletal issues and stress ([SF Søndergaard 2022](#)). There is a small but statistically significant increase of 1% in the percentage of patients reporting a positive experience.

Limitations and discussion

Data quality and availability

Our analyses rely on historical data collections, some of which do not go back as far as 2008 (see [Figure 1](#)). There are also gaps/anomalous data in the time series for some providers and potential control sites. How some metrics have been measured is likely to have changed over time, but the causal impact analysis uses control data from sites without interventions which will help to correct for any changes occurring at all sites. Some study sites have been involved in trust mergers/site changes in the time period prior to switching to single bedrooms; we have merged historic data where appropriate, but these mergers may have impacted the time series.

Some variables are only available at provider level ([Table 3](#)), so where providers have more than one site any change that may occur following a switch to single bedrooms at one site will be diluted by a lack of change at the providers other sites.

The variables that we have measured are not always entirely specific to the impact we are trying to measure, for example we are using both staff sickness and staff turnover as a proxy for staff experience, with the assumption that increased staff turnover and sickness absences could indicate that staff are less happy in their workplace following a switch to single bedrooms. It is possible that sickness absences may be influenced in other ways by a switch to single bedrooms, for example if single bedrooms reduce outbreaks of contagious illnesses such as influenza and better protect staff from contracting them themselves, then sickness absences could be reduced. Indeed, the staff and patient surveys will be influenced by many other factors not just single bed rooms. Median waiting time is used as a proxy for how productive a hospital is, but waiting times will depend on not only the flow of patients through the hospital, but the numbers being referred. If a new hospital gets a large increase in referrals then even with an moderate increase in productivity there could still be an increase in waiting times as the waiting list size grows.

Some measures are recorded quarterly (bed occupancy) or annually (cleaning costs and staff survey) so we have less data points with which to fit our models and assess outcomes, and less temporal specificity to detect changes.

COVID-19 has impacted many of the variables for sites that switched just before or after the onset of the pandemic (Royal Liverpool, Clatterbridge and Papworth). While the causal impact analysis does use control sites to account for changes over time that would have happened without the intervention, the magnitude of the impact of the COVID-19 was sometimes different between study sites and their controls. Papworth, in particular, appeared to be more impacted by COVID-19 than its control sites.

Model fit and methodology

Model fit was assessed using the Mean Absolute Percentage Error (MAPE) and Durbin-Watson statistic, and adjustments were made to try to ensure a good historical model fit while minimising the forecasting intervals. However, for some data the models will be a less good fit and will have large credible intervals in the outcome period due to the variability in the indicator not being well explained by the control time series.

Three of the seven study sites were specialist sites (Papworth is a specialist cardiothoracic/transplant centre, Clatterbridge is a cancer centre and Chase Farm only conducts elective care), making it more difficult to select well matched control sites. The number of sites in the control pool was very limited for both Clatterbridge (4) and Papworth (9), which may impact the quality of the model fit and reliability of the outcomes.

The three specialist provider sites were excluded from the meta-analysis as the impacts on these sites may be less generalisable to new general acute hospitals that may be built in the future. The meta-analyses were weighted based on the inverse variance of the relative effects, but other factors that may influence how much weight should be given to a particular site were not included.

General

For our analysis we have had to specify the month in which the switch to single bedrooms occurred. However, at some sites the switch was done over a longer period of time with certain services moving at different stages. For example, Tunbridge Wells moved Women and Children's wards to single bed rooms in January 2011 and the

remainder of inpatient services in September 2011. We have used the month when the switch started, so that control site selection and predictions are not affected by changes in the transition period, but this means that we may see initial changes related to the transition period or that some changes may not be seen until later in the outcome period when the full switch had occurred.

When the sites switched to single bedrooms there are likely to have been other changes to how services operate and at multi-site providers there may also have been changes to what services are provided at which sites, so there are likely to be confounding factors that influence our results.

Not all sites switched entirely to single bed rooms and all sites had differing proportions of single bedrooms prior to the switch. There does not appear to be a clear relationship between the percentage of single bedrooms post switch or the change in proportion of single bed and the varying outcomes that are seen at individual sites. The sites that switched to single bedrooms will have had different designs and the differences in results between sites may be due to different single bedroom designs/implementation working better, or worse, than others. It is also possible that the best design/implementation may vary by hospital type so that a design that suits a general acute hospital may work less well at a specialist site. Where there are potential negative impacts of a switch to single bedrooms some sites may have had mitigating factors in place from the start while other schemes may have brought in mitigating factors at a later date once issues had already arisen.

We see some limited evidence that the rate of hospital acquired C. difficile infections may be reduced, but unfortunately it was not possible to assess other healthcare acquired infections (MSSA, MRSA, Klebsiella and P. aeruginosa) due to a lack of data availability before 2017. In addition, there is no data on other hospital acquired contagious illnesses, such as influenza and COVID-19, where the spread may be reduced by single bed rooms. It is possible that due to en suite bathrooms in single bedroom hospitals, that there is an increased risk of Legionella, due to increased pipework and bathrooms potentially not all being used regularly.

The following table are a subjective rating of our confidence in the indicators to assess the impacts of SBR given the above caveats and limitations:

Further notes on data quality and reasons for exclusion of sites from indicator analysis are in [Section 12.2](#).

Recommendations for modelling and future evaluation

While the availability of data has improved since 2008, there is still a need to ensure consistent and accurate data recording at ideally a site level, in order to more accurately evaluate the impact of any future hospital designs.

Given the apparent negative impacts on staff of a switch to single bedrooms it is likely important going forward that more data is collected on the views of staff specifically around switching to single bedrooms, including qualitative data to better understand what concerns and difficulties there may be.

Also, there is more data is needed to fully understand the impacts of single bedrooms on infection control, including things such as hospital acquired influenza or COVID-19.

Further analysis would be needed to fully understand the impact on waiting times and waiting list sizes, given the dynamic nature of waiting lists. This is a measure where there could be cumulative effects over time, and also a

new or redesigned hospital could result in more or less people being referred, so waiting list size/time will not just be dependent on hospital productivity.

Future work should focus on understanding how/whether outcomes in single room hospitals may be impacted by factors such as the layout and design of a hospital, culture, use of technology and ways of working.

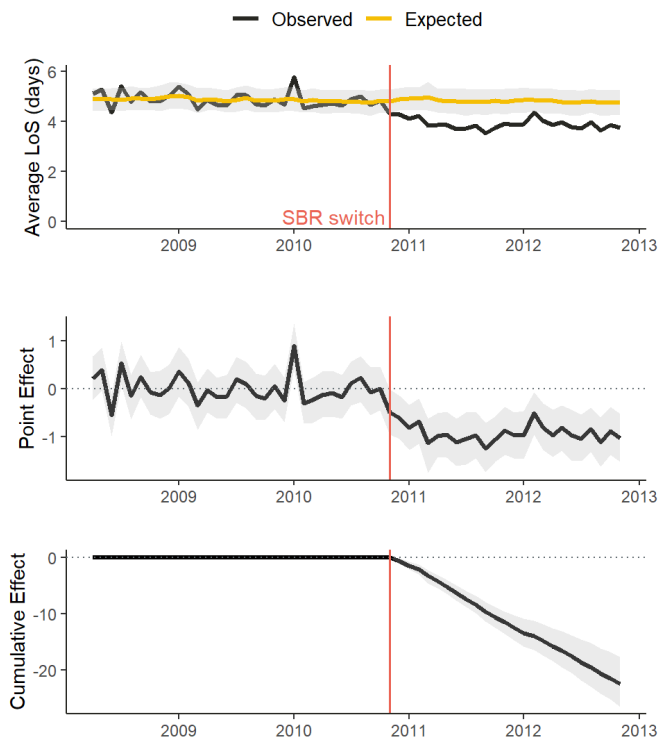
Appendices

A. CIA charts by indicator and hospital site/provider

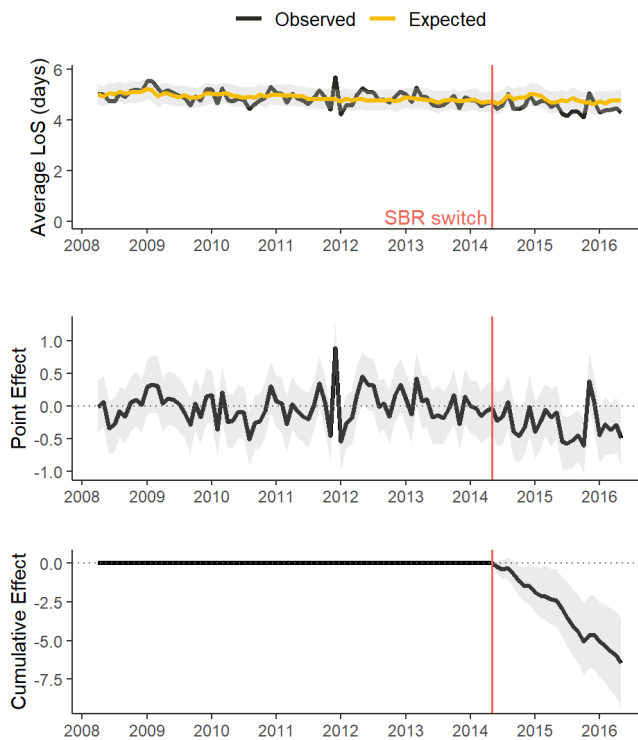
Length of stay

Decrease	No change	Increase
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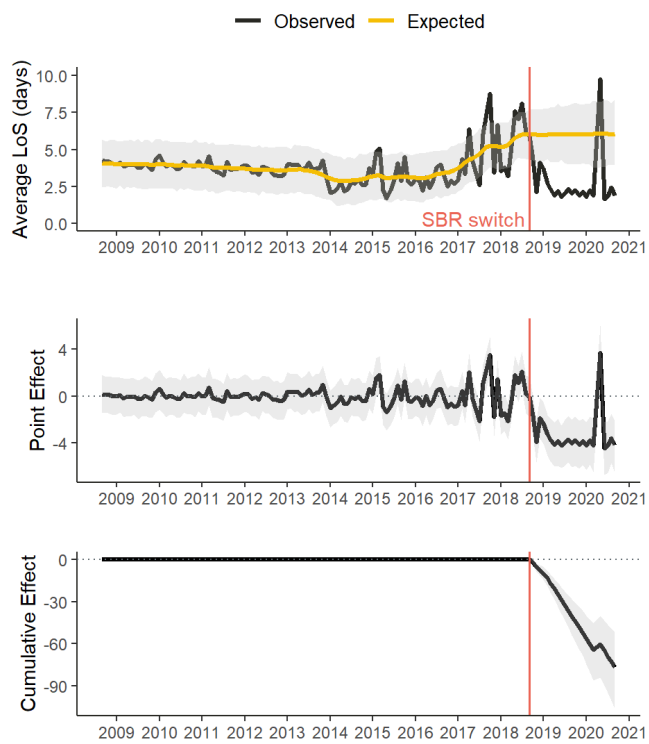
Peterborough



Southmead



Chase Farm



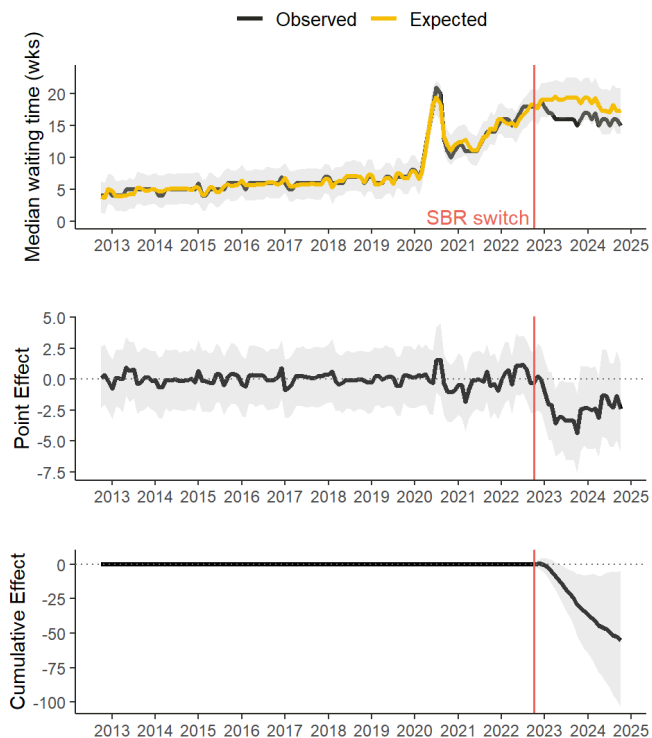
Median waiting time

Decrease

No change

Increase

Royal Liverpool



Peterborough

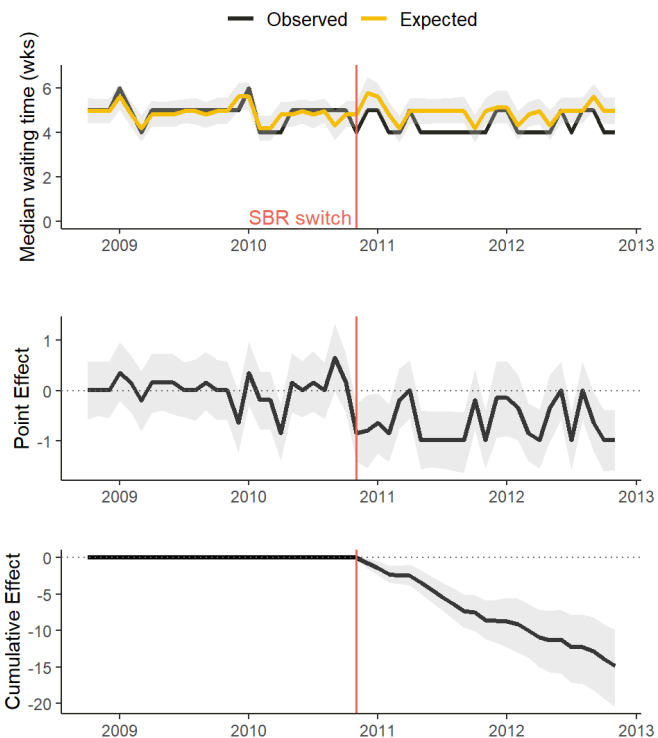


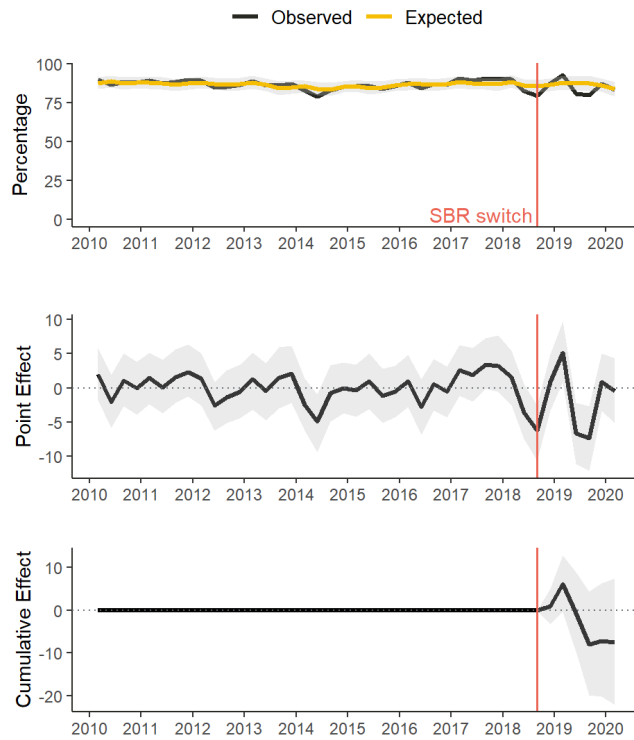
Figure 3: ?(caption)

Bed occupancy

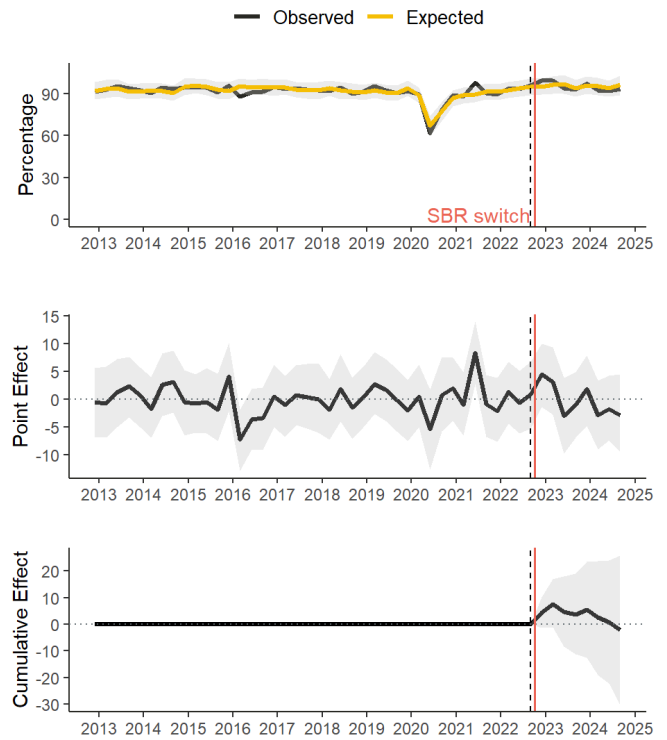
No change

Increase

Chase Farm



Royal Liverpool

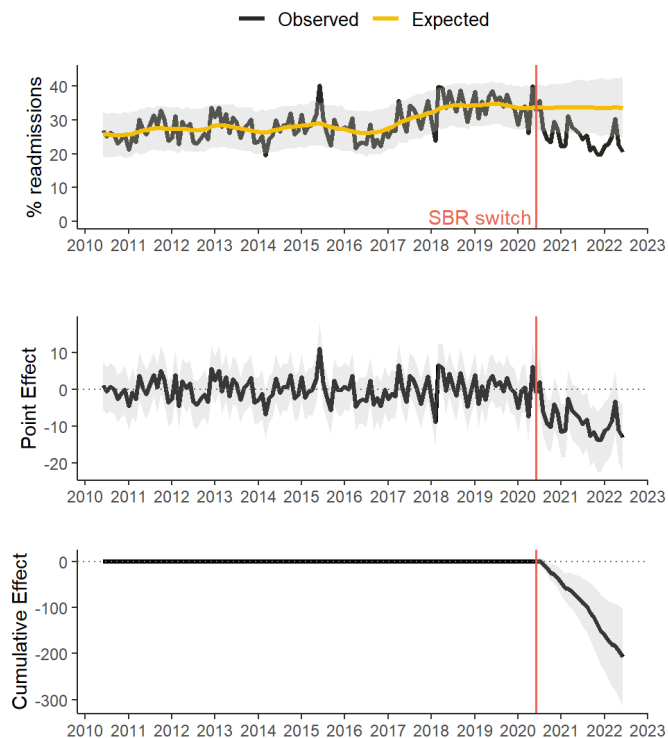


Emergency readmissions

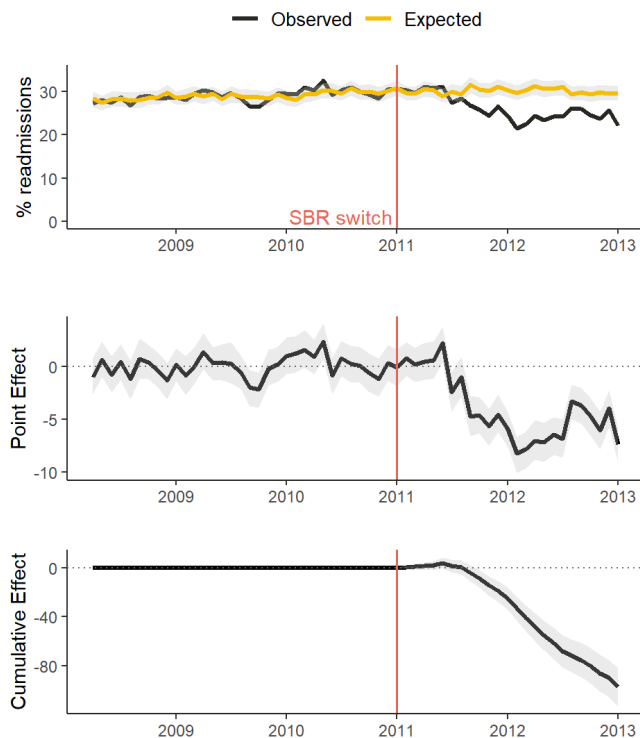
Decrease

No change

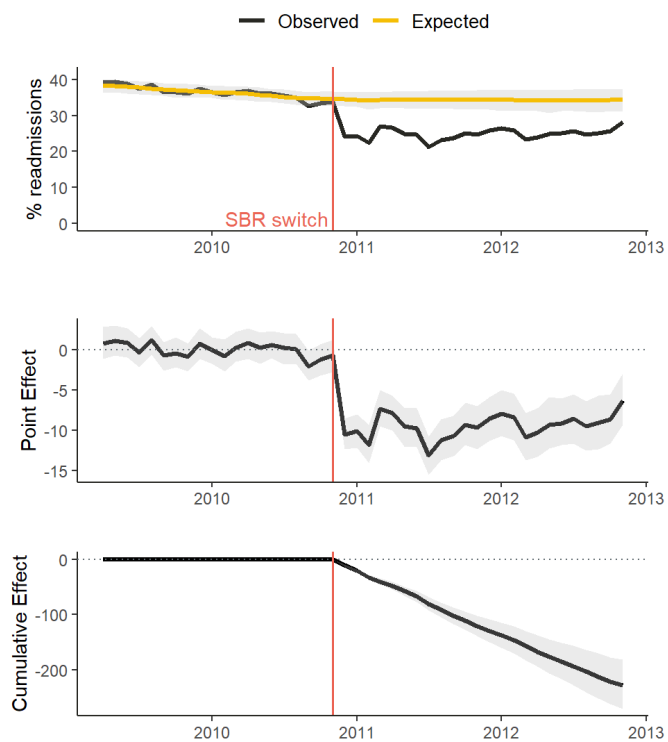
Clatterbridge



Tunbridge Wells



Peterborough

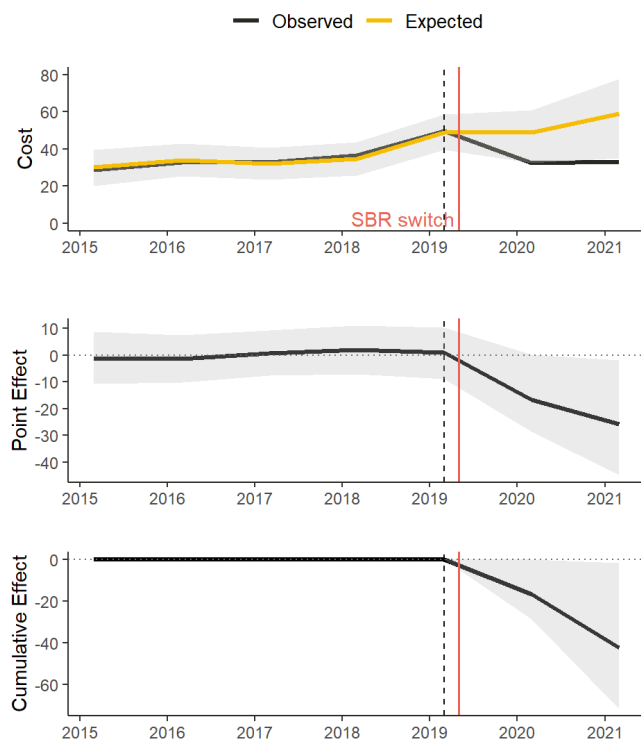


Cleaning costs

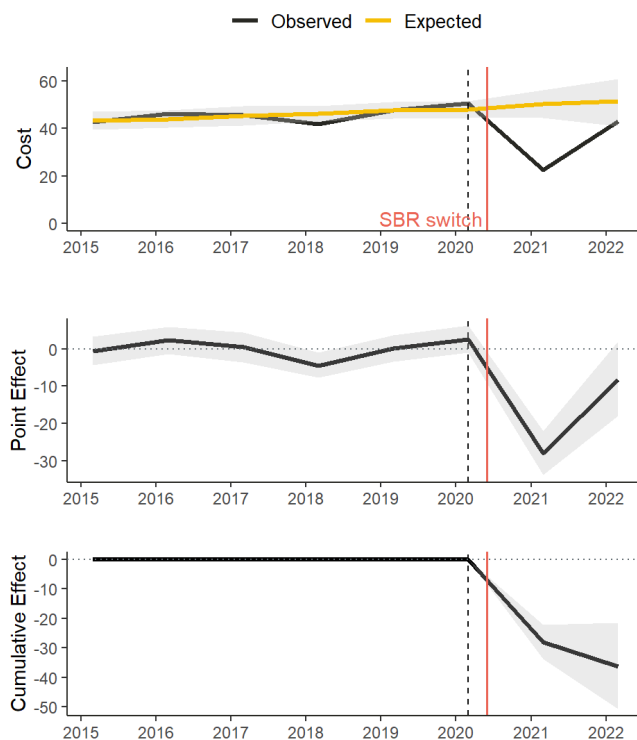
Decrease

No change

Papworth (RGM)



Clatterbridge (REN)



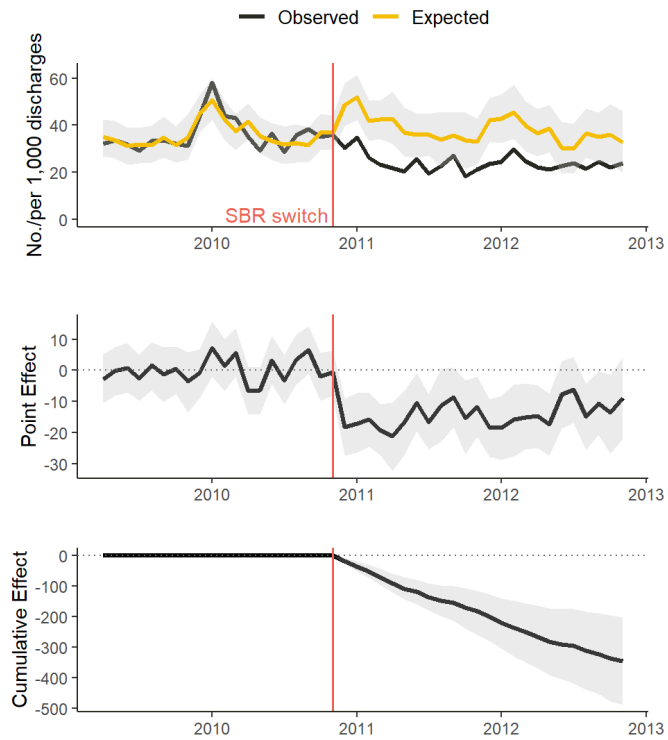
Black dotted line indicates the time of the last data point prior to the switch to single bedrooms

Deaths in hospital

Decrease

No change

Peterborough

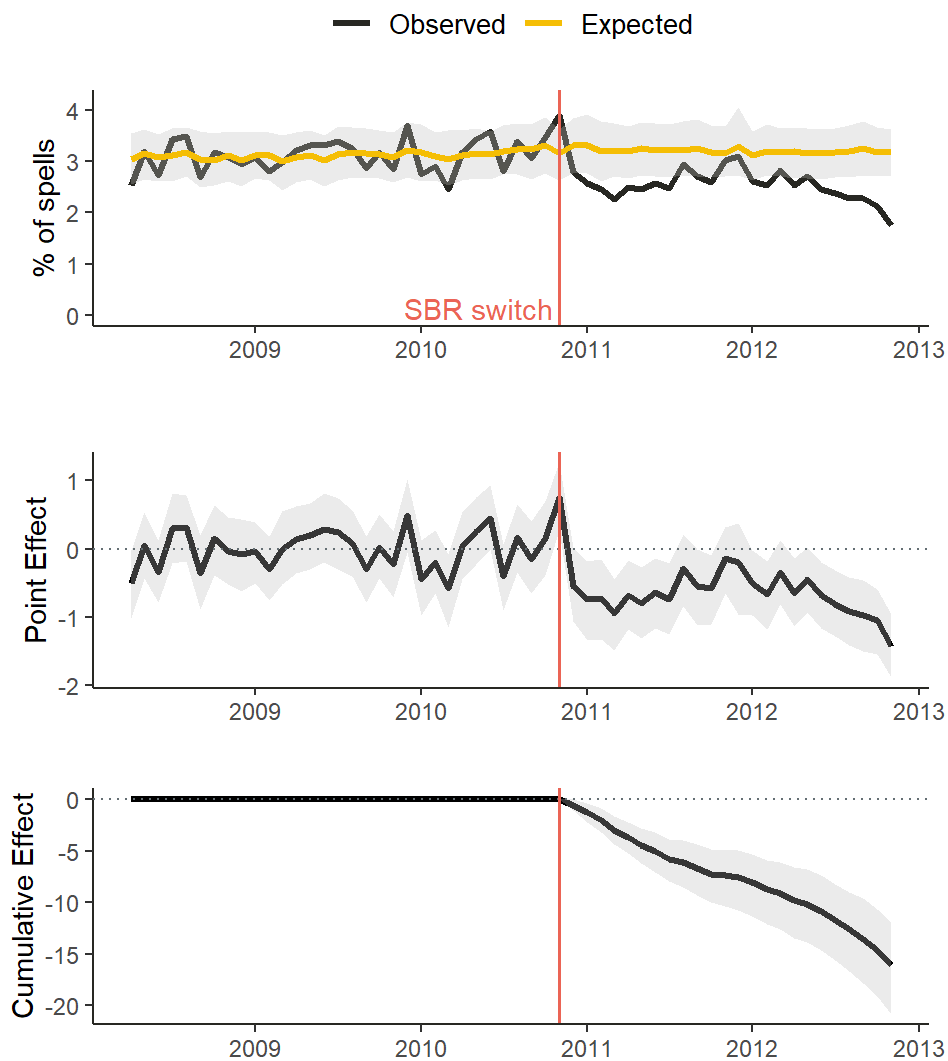


Falls and injuries in hospital

Decrease

No change

Peterborough

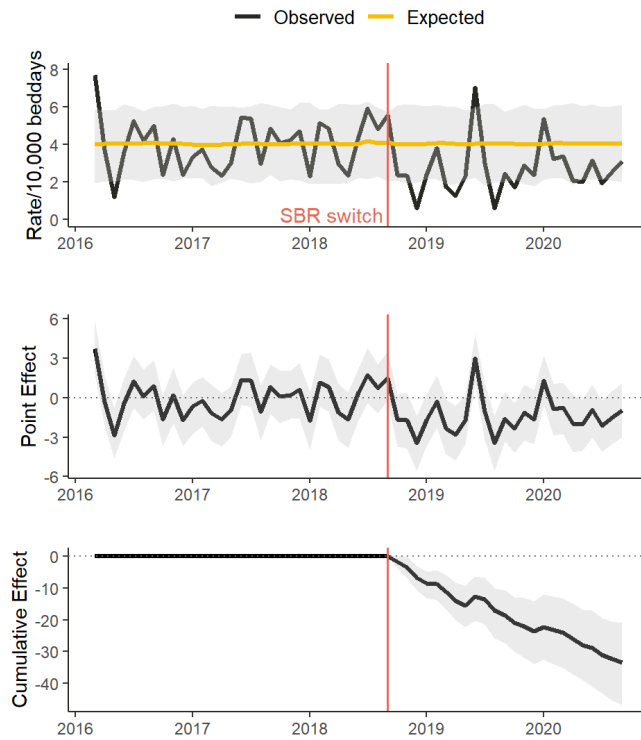


Healthcare acquired c.difficile infections

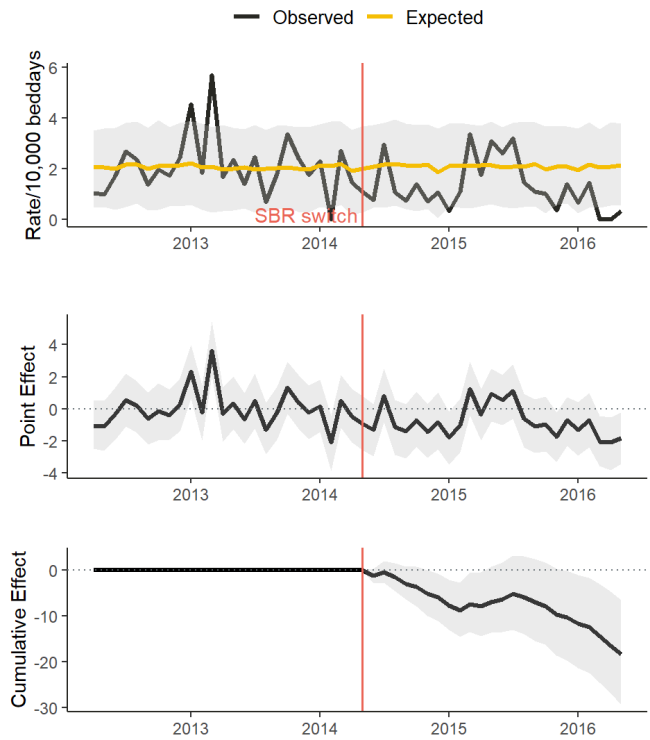
Decrease

No change

Chase Farm



Southmead



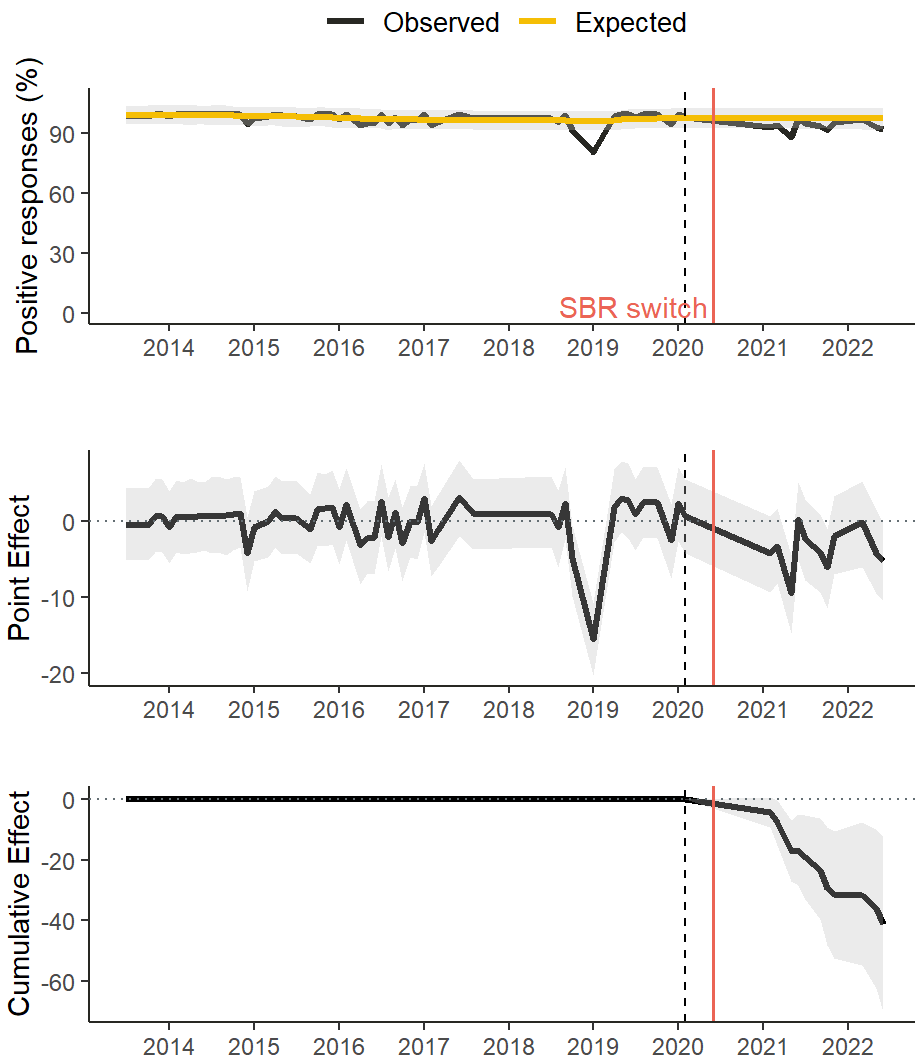
Patient experience - Friends and Family Test

Decrease

Increase

No Change

Clatterbridge (REN)

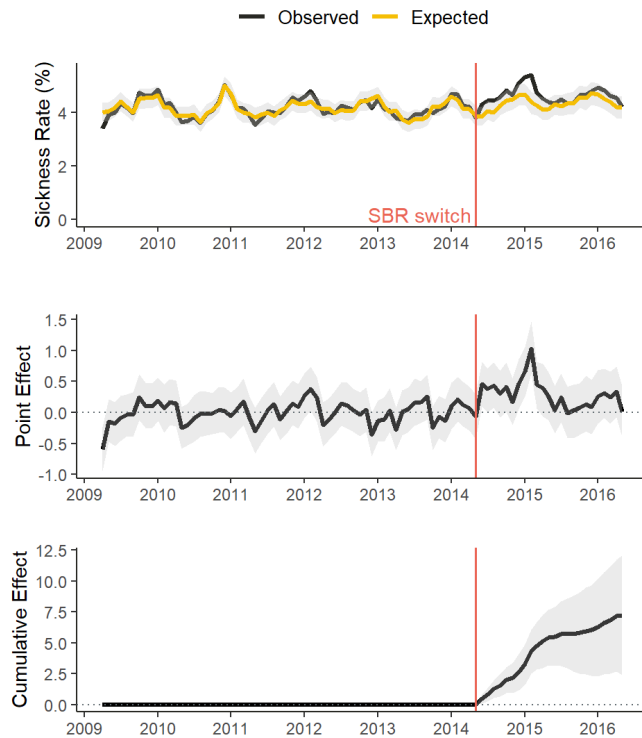


Staff sickness

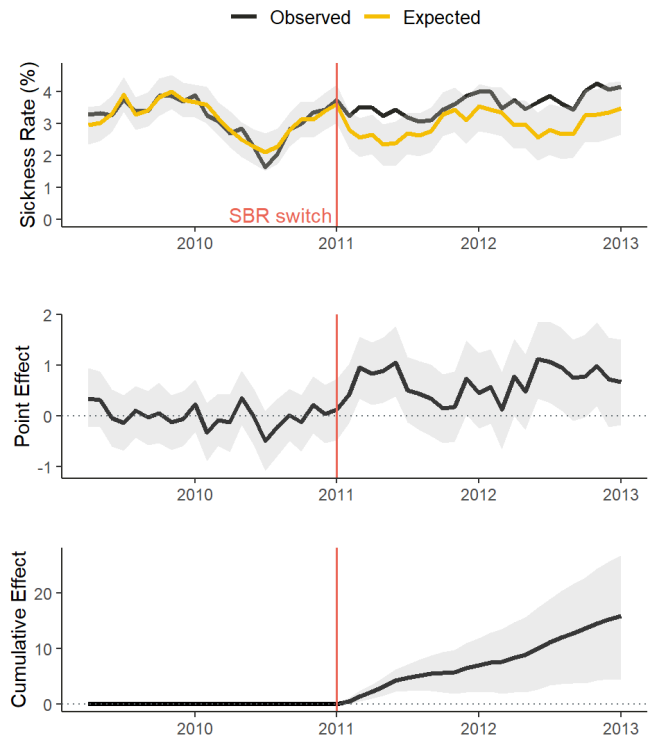
Increase

No change

Southmead (RVJ)



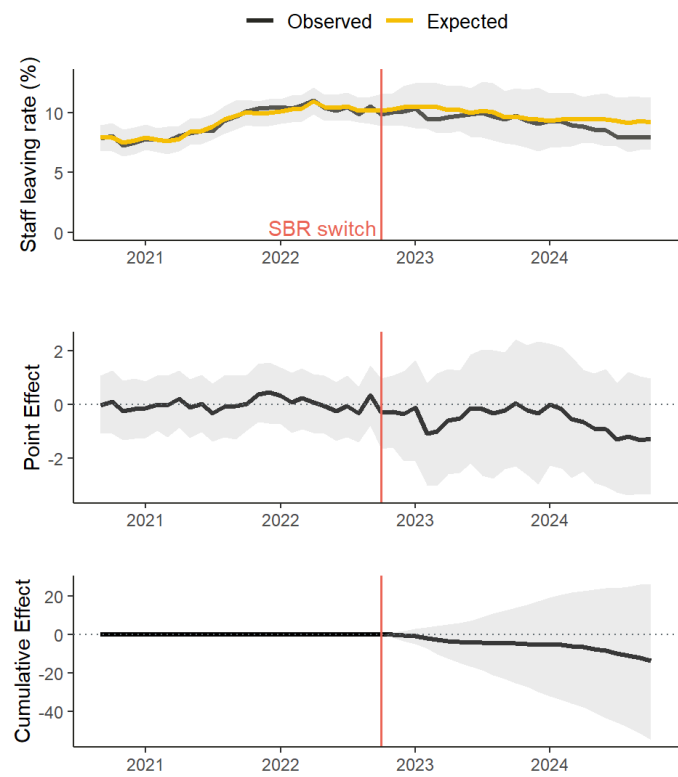
Tunbridge Wells (RWF)



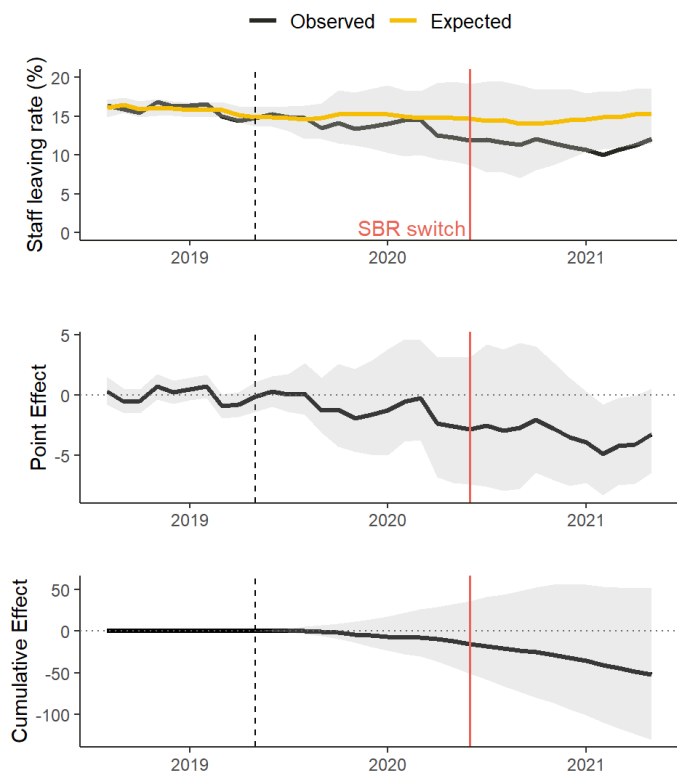
Staff turnover

No change

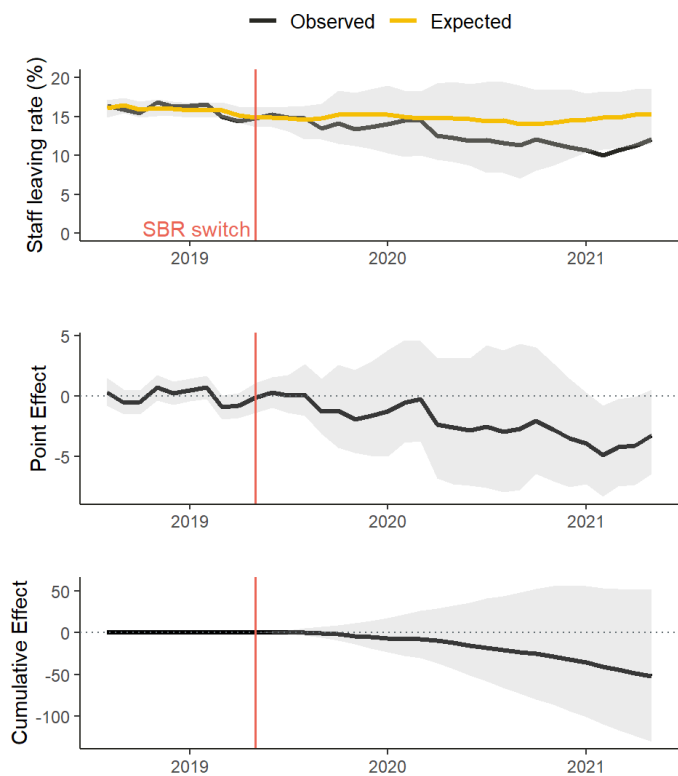
Royal Liverpool (REM)



Clatterbridge (REN)



Papworth (RGM)



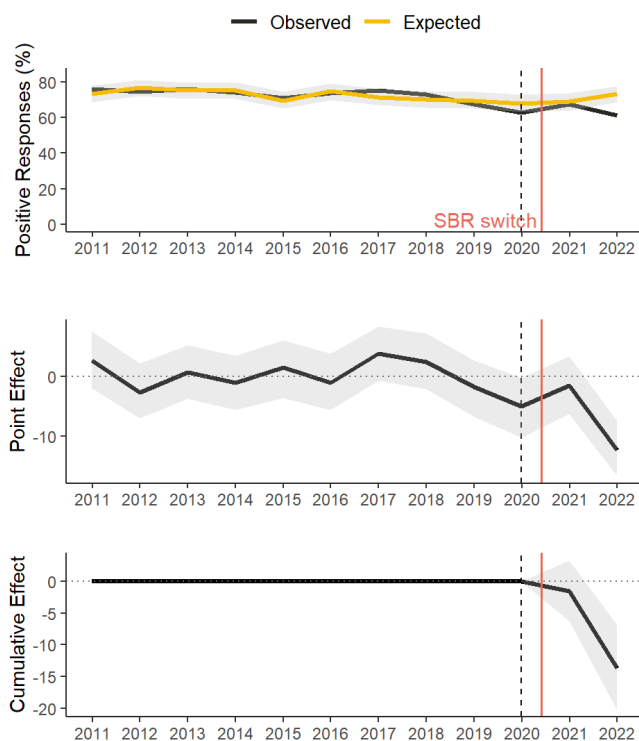
Staff survey

Decrease

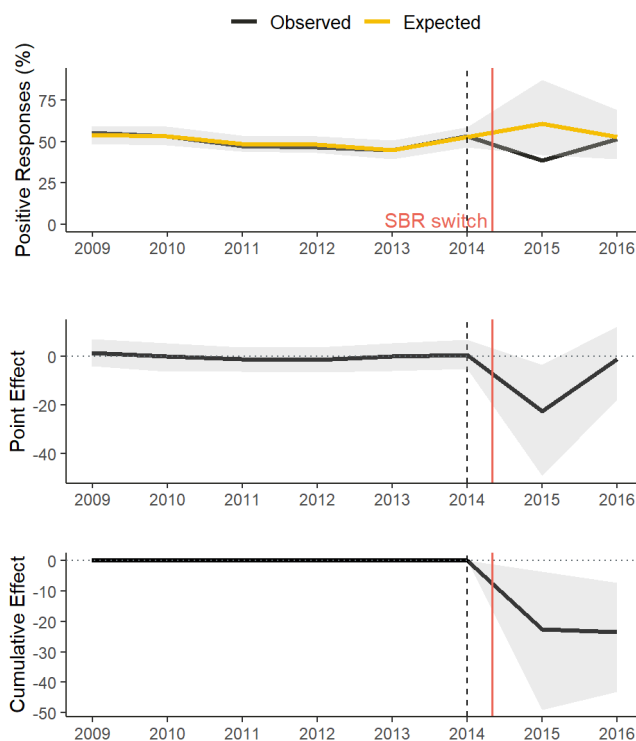
No change

Increase

Clatterbridge (REN)



Southmead (RVJ)



Black dotted line indicates the time of the last data point prior to the switch to single bedrooms

B. Notes on data quality and suitability (for CIA)

Referral to treatment waiting time data

- Only available at provider level.
- This measure is not specific to waiting time for inpatient treatment that will be most impacted by a switch to single bedrooms, as it will include those waiting for diagnostics and for non-admitted treatment.
- Chase Farm was excluded as there was only 4 months of data available in the outcome period.
- Southmead was excluded due to a period of missing data in the 2 years prior to the switch to single rooms.

Length of stay

- Papworth was excluded as there were 2 large COVID-19 related spikes in length of stay in the outcome period that were significantly larger than the COVID-19 effects in the control sites, so it was not possible to determine if there would have been a change in length of stay without COVID-19 and thus this data has been excluded from the analysis.

Bed occupancy

- Only available at provider level.
- This is a quarterly measure so there is more limited data and depending on the time point at which the switch to single bedrooms occurred there may be no data until 3 months after the switch or the last point of baseline may be 3 months prior to the switch.
- For Papworth bed occupancy was highly affected by COVID-19 in the outcome period, an effect which was not seen to the same extent at the control sites, so it wasn't possible to determine whether there would have an effect on bed occupancy had COVID-19 not occurred.

Cleaning costs

- This is an annual measure, so there are limited data points in the pre-switch period and only 1 or 2 points in the outcome period.
- There was missing occupied floor space data for 2020, 2021 and 2024, so we have imputed values based on the available data and the time point when the switch to single bedrooms occurred.
- This measure is calculated as cost per occupied square metre, but is not specific to inpatient bed areas, but includes all other clinical areas within the hospital, e.g. operating theatres, outpatient clinics etc.
- Meta-analysis is not possible as Royal Liverpool is the only general acute hospital for which we have data.
- For Chase Farm the cleaning costs were quite variable and it was difficult to find appropriate controls, so we were not able to fit a suitable model.

Healthcare acquired infections

- Only available at provider level.
- C. difficile infection rates could not be analysed for Clatterbridge or Papworth as the rates were low, often zero, and so combined with the high level of variability in the data it meant it was not possible to fit a suitable model for these sites.
- For other hospital acquired infections the data collections did not go far enough back in time to be used in this analysis.

Falls and injuries in hospital

- Clatterbridge, Papworth and Chase Farm were all excluded as the rates were very low, often zero, so it wasn't possible to fit suitable models to the data.

Deaths in hospital

- Papworth was excluded due to a large COVID-19 spike in the outcome period that was greater than that seen in the control sites, so it was not possible to determine if there would have been a change had COVID-19 not occurred.
- Chase Farm was excluded as the rate was very low, often zero, so it wasn't possible to fit a suitable model, this low rate is unsurprising given it is only conducting elective care.

Patient experience

- There is around a year of missing Friends and Family Test data for all sites during the COVID-19 pandemic (March 2020 until late 2020/early 2021), this means that there is some missing data in the baseline period for

Royal Liverpool and some missing data in the outcome periods for Clatterbridge, Papworth and Chase Farm.

Staff sickness

- Only available at provider level.
- This is used as a proxy for staff experience with the assumption that sickness absences would rise if staff were stressed and unhappy in their roles.

Staff turnover

- Only available at provider level.
- Meta-analysis is not possible as Royal Liverpool is the only general acute hospital for which we have data.
- This is used as a proxy for staff experience with the assumption that the staff leaving rate may increase if staff are unhappy in their roles.

Staff survey

- Only available at provider level.
- This is an annual measure, so there are limited data points in the pre-switch period and only 1 or 2 points in the outcome period.

C. Control pools for each intervention site

As outlined in [Section 8.1](#), there was a multi-phase control selection process to create a pool of control sites for each intervention site for further analysis. The most similar control sites for each intervention site are shown below. Note that, whilst controls will always be selected from within these sets, the exact controls will not be the same for every indicator as the CIA approach further refines matches based on similar pre-intervention time series.

Royal Liverpool

Clatterbridge

Royal Papworth

Peterborough

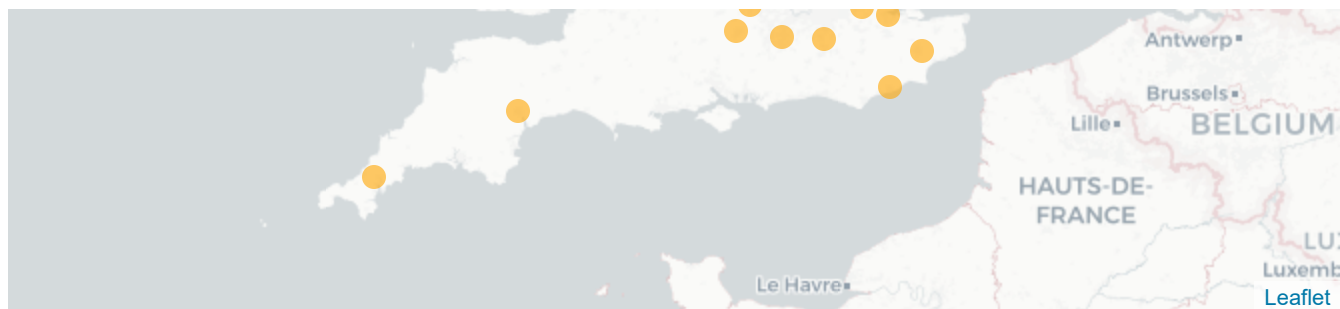
Chase Farm

Southmead

Tunbridge Wells

Map of Royal Liverpool and similar control sites





Provider	Provider code	Site	Site code
Stockport NHS Foundation Trust	RWJ	Stepping Hill Hospital	RWJ09
Royal Devon University Healthcare NHS Foundation Trust	RH8	Royal Devon & Exeter Hospital (Wonford)	RH801
Royal Berkshire NHS Foundation Trust	RHW	Royal Berkshire Hospital	RHW01
Medway NHS Foundation Trust	RPA	Medway Maritime Hospital	RPA02
Dartford and Gravesham NHS Trust	RN7	Darent Valley Hospital	RN707
East Sussex Healthcare NHS Trust	RXC	Conquest Hospital	RXC01
Royal Cornwall Hospitals NHS Trust	REF	Royal Cornwall Hospital (Treliske)	REF12
Worcestershire Acute Hospitals NHS Trust	RWP	Worcestershire Royal Hospital	RWP50
Bedfordshire Hospitals NHS Foundation Trust	RC9	Bedford Hospital South Wing	RC979
Sandwell and West Birmingham Hospitals NHS Trust	RXK	Sandwell General Hospital	RXK01
Sandwell and West Birmingham Hospitals NHS Trust	RXK	City Hospital	RXK02
Lancashire Teaching Hospitals NHS Foundation Trust	RXN	Royal Preston Hospital	RXN02
Northampton General Hospital NHS Trust	RNS	Northampton General Hospital (Acute)	RNS01
Barking, Havering and Redbridge University Hospitals NHS Trust	RF4	Queen's Hospital	RF4QH
Hampshire Hospitals NHS Foundation Trust	RN5	Basingstoke and North Hampshire Hospital	RN506
Kettering General Hospital NHS Foundation Trust	RNQ	Kettering General Hospital	RNQ51
Surrey and Sussex Healthcare NHS Trust	RTP	East Surrey Hospital	RTP04
Royal Surrey County Hospital NHS Foundation Trust	RA2	Royal Surrey County Hospital	RA201
West Suffolk NHS Foundation Trust	RGR	West Suffolk Hospital	RGR50
East Kent Hospitals University NHS Foundation Trust	RVV	William Harvey Hospital (Ashford)	RVV01

n: 20

D. List of control sites/providers by indicator

A different set of controls was identified by [MarketMatching](#) for each indicator and each intervention hospital for the CIA ([Section 8.2](#)). A full list of the sites or providers used in each CIA is available below for each indicator and intervention hospital.

Length of stay Median waiting time Bed occupancy Emergency readmissions

Cleaning costs Deaths in hospital Falls and injuries in hospital

Healthcare acquired infections - combined rate Healthcare acquired c.difficile infections

Patient experience - Friends and Family Test Staff sickness Staff turnover Staff survey

Provider	Provider code	Site	Site code
Chase Farm Hospital			
Calderdale and Huddersfield NHS Foundation Trust	RWY	Huddersfield Royal Infirmary	RWY01
Doncaster and Bassetlaw Teaching Hospitals NHS Foundation Trust	RP5	Bassetlaw Hospital	RP5BA
Hampshire Hospitals NHS Foundation Trust	RN5	Basingstoke and North Hampshire Hospital	RN506
Harrogate and District NHS Foundation Trust	RCD	Harrogate District Hospital	RCD01
Pennine Acute Hospitals NHS Trust	RW6	Fairfield General Hospital	RW601
Clatterbridge Cancer Centre			
Sheffield Teaching Hospitals NHS Foundation Trust	RHQ	Weston Park Hospital	RHQWP
The Christie NHS Foundation Trust	RBV	The Christie	RBV01
The Newcastle Upon Tyne Hospitals NHS Foundation Trust	RTD	Northern Centre For Cancer Care	RTD06
University Hospitals Bristol and Weston NHS Foundation Trust	RA7	Bristol Haematology & Oncology Centre	RA710
Peterborough (city) Hospital			
Ashford and St Peter's Hospitals NHS Foundation Trust	RTK	St Peter's Hospital	RTK01
Brighton and Sussex University Hospitals NHS Trust	RXH	Royal Sussex County Hospital	RXH01
East Suffolk and North Essex NHS Foundation Trust	RDE	Colchester General Hospital	RDEE4
Northampton General Hospital NHS Trust	RNS	Northampton General Hospital (Acute)	RNS01
St George's University Hospitals NHS Foundation Trust	RJ7	St George's Hospital (Tooting)	RJ701
Royal Liverpool			

Provider	Provider code	Site	Site code
Barking, Havering and Redbridge University Hospitals NHS Trust	RF4	Queen's Hospital	RF4QH
Kettering General Hospital NHS Foundation Trust	RNQ	Kettering General Hospital	RNQ51
Royal Devon University Healthcare NHS Foundation Trust	RH8	Royal Devon & Exeter Hospital (Wonford)	RH801
Sandwell and West Birmingham Hospitals NHS Trust	RXK	Sandwell General Hospital	RXK01
Surrey and Sussex Healthcare NHS Trust	RTP	East Surrey Hospital	RTP04
Royal Papworth			
Leeds Teaching Hospitals NHS Trust	RR8	Leeds General Infirmary	RR801
Leeds Teaching Hospitals NHS Trust	RR8	St James's University Hospital	RR813
Nottingham University Hospitals NHS Trust	RX1	Nottingham University Hospitals NHS Trust - City Campus	RX1CC
University Hospitals Bristol and Weston NHS Foundation Trust	RA7	Bristol Royal Infirmary	RA701
University Hospitals Of North Midlands NHS Trust	RJE	Royal Stoke University Hospital	RJE01
Southmead Hospital			
East Suffolk and North Essex NHS Foundation Trust	RDE	Colchester General Hospital	RDEE4
Ipswich Hospital NHS Trust	RGQ	The Ipswich Hospital NHS Trust	RGQ02
Mid and South Essex NHS Foundation Trust	RAJ	Southend Hospital	RAJ01
Royal Berkshire NHS Foundation Trust	RHW	Royal Berkshire Hospital	RHW01
Royal Surrey County Hospital NHS Foundation Trust	RA2	Royal Surrey County Hospital	RA201
Tunbridge Wells			
Mid Staffordshire NHS Foundation Trust	RJD	Stafford Hospital	RJD01
Sandwell and West Birmingham Hospitals NHS Trust	RXK	Sandwell General Hospital	RXK01
Sandwell and West Birmingham Hospitals NHS Trust	RXK	City Hospital	RXK02
Warrington and Halton Teaching Hospitals NHS Foundation Trust	RWW	Warrington Hospital	RWWWH
West Hertfordshire Teaching Hospitals NHS Trust	RWG	Watford General Hospital	RWG02

E. Single Bedroom Percentage

As mentioned in [Section 6](#), data on the percentage of single bedrooms was not available at site level. However at provider level, the single bedroom percentage was calculated as the number of single bedrooms according to the annual March ERIC dataset ([NHS England 2023](#)) over the number of available beds in the March quarterly KH03 return. The switch to single bedrooms was at site level, but since the KH03 data is only available at provider level, the single bedroom percentage is at provider level. So at providers with multiple sites, the single bedroom percentage may appear lower than expected. Despite this, these charts are useful at demonstrating that the single bedroom percentage did change over time at the study sites compared to the control pool (in grey) and give the best indication of the scale of the change in single bedroom percentage that the data allows.

Royal Liverpool

Clatterbridge

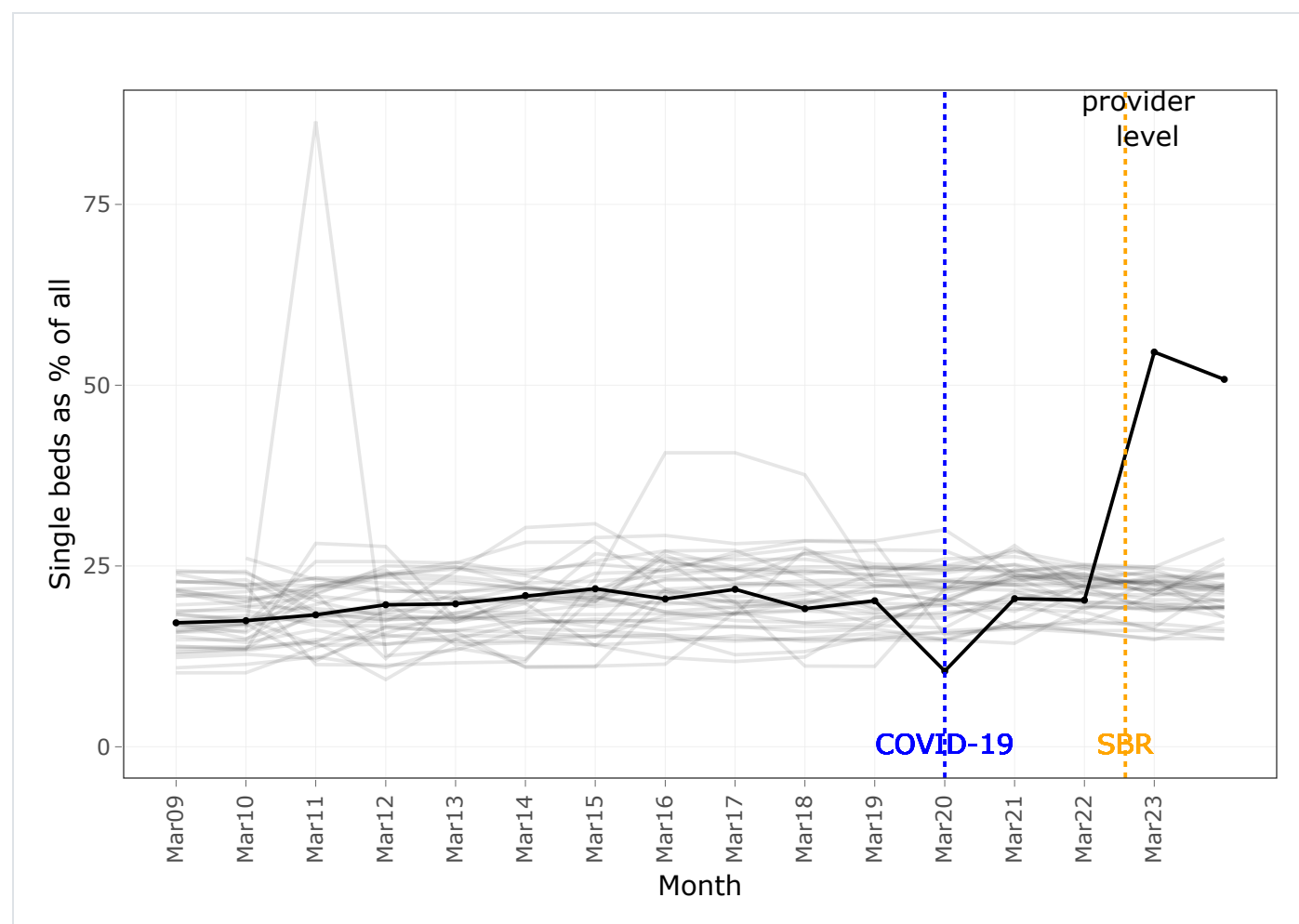
Royal Papworth

Peterborough

Chase Farm

Southmead

Tunbridge Wells



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